Draft version July 30, 2026
Typeset using LaTeX **twocolumn** style in AASTeX631

Dark-siren inference of H_0 and the AGN-hosted fraction: what a sparse, strongly clustered tracer adds, and what the host survey must deliver

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(Received July 30, 2026)

ABSTRACT

Statistical host identification turns a gravitational-wave distance into a cosmological measurement by borrowing redshifts from a galaxy catalog, and it is usually done with the densest catalog available. We ask what a second, much sparser and more strongly clustered tracer buys: whether the fraction of compact-binary mergers hosted by active galactic nuclei (AGN) can be measured jointly with the expansion rate, and what a real host survey would have to deliver for that measurement to survive. Using clustered lognormal mock catalogs with a galaxy–AGN bias contrast of 1.2 against 2.0 and a comoving number-density ratio of 100, we recover the AGN-hosted fraction across its entire range from 1 000 dark sirens, with the planted value inside the 68% interval at every interior point and residual offsets no larger than 0.025. Taking a matched two-tracer survey from complete down to 17.5% completeness within the detection horizon degrades $\sigma(f_{\mathrm{AGN}})$ by only 1.11×, while $\sigma(H_0)$ degrades by up to 1.87× — non-monotonically, and with the first rung nearly free, because the hosts a flux limit removes first lie beyond the horizon. That robustness, however, is bought entirely by knowing the sparse tracer's number density. Freeing it opens a strong degeneracy with the mixture weight (ρ up to +0.90) and the detection significance of f_{AGN} falls from 8.2σ to 1.2σ; anchoring the AGN density to 0.041 dex — about 10% — costs almost nothing at any completeness, so density knowledge, not survey depth, is the binding requirement. We close with an error budget of the distance scale that closes end to end. The clustered mocks' H_0 deficit is the incomplete cancellation of two spurious levers sprung by a single inconsistency — a true-redshift detection cut expressed as a distance-space selection against catalogs with support above the horizon — and an estimator matched to the detection rule recovers the truth; the deficit's growth with f_{AGN} is the sparse tracer acting as a distance anchor, a genuinely multitracer robustness. On a matched mock, the offset decomposes into two generator defects — detection acting on latent data (+0.768±0.226 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$) and a sky-localisation width derived from latent parameters (−0.49±0.08, a defect that biases even the exact likelihood) — plus an estimator overhead of −0.31±0.13. Regenerating the mocks with the repaired generator closes the budget end to end: the recovered expansion rate moves to −0.39±0.15 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ over twenty single-tracer realisations — the predicted estimator overhead, not a defect — and the twelve-realisation two-tracer ensemble recovers H_0 at +0.44±0.36 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Independently, the host-catalog realisation contributes 0.97 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ of scatter per 1 000-event realisation, which single-catalog credible intervals understate by a factor of 2.2.

Keywords: Gravitational waves — Cosmology — Hubble constant — Active galactic nuclei — Galaxy catalogs

1. INTRODUCTION

A compact-binary merger measures its own luminosity distance and nothing about its redshift. The dark-siren programme supplies the missing redshift statistically: instead of one identified host, the analysis uses the redshift distribution of all catalogued galaxies consistent with the event's sky localisation, and marginalises over which of them was the host (Schutz 1986; Del Pozzo 2012). With enough events the distance–redshift relation, and hence the expansion rate, is recovered without a single electromagnetic counterpart (Abbott & et al. 2023a; Abac & et al. 2025).

The precision of that inference is set by the catalog, not by the gravitational-wave data alone. Two catalog properties do the work. The first is how tightly the catalog pins the redshift of a host given the event's sky patch: a dense catalog with good redshifts turns a wide distance posterior into a narrow set of candidate redshifts, and the information scales with how structured — how clustered — that redshift distribution is on the scale of the localisation (Fishbach et al. 2019; Gray et al. 2020). The second is how well the missing galaxies are accounted for. Real catalogs are flux limited, so beyond some redshift most hosts are absent, and the analysis must either restrict itself to the complete regime or model the missing population (Chen et al. 2018; Finke et al. 2021; Gray et al. 2022). Both properties are properties of a survey, and both are usually satisfied by choosing the densest galaxy catalog available.

There is, however, a reason to look at a second and much sparser tracer. Several channels predict that a subset of binary black hole mergers occurs in the gas-rich, dense environments around active galactic nuclei (Bartos et al. 2017; Stone et al. 2017; McKernan et al. 2018; Tagawa et al. 2020), and the fraction of the merger population that this channel supplies, $f_{\rm AGN}$, is a population parameter with direct astrophysical content: it is the branching ratio between an AGN-assisted assembly channel and everything else. It is also, in principle, measurable from the same data that measure the expansion rate, because AGN are not distributed like galaxies. An AGN catalog is sparse — in the mocks used here its comoving number density is lower by a factor of 100 — and it is more strongly clustered, with a linear bias of 2.0 against 1.2 for the galaxies, a contrast of 1.67. Sparsity and bias act in opposite directions: sparsity means most sky patches contain no AGN at all, which is informative precisely because an AGN-hosted event cannot have come from an empty patch, while the higher bias means the AGN that do exist trace the same structures more sharply. The joint inference therefore has a channel that no single-tracer analysis has: the *contrast* between two host priors built on the same sky.

Whether that channel is usable is an empirical question, and it is the question this paper answers. We work entirely in simulation, on mocks whose ingredients are under our control, and we ask three things in order. Can the AGN-hosted fraction be recovered jointly with the expansion rate, and over what range? How fast does that measurement degrade as the host survey becomes realistically incomplete? And what does the analysis have to be told about the sparse tracer — specifically, about its comoving number density — for the answer to survive?

The answers are, in short: yes over the full range of $f_{\rm AGN}$; slowly with incompleteness; and a lot. The AGN-hosted fraction is recovered across its entire planted range from 1 000 dark sirens on complete clustered catalogs (§3). Taking a two-tracer survey from complete down to 17.5% completeness within the detection horizon — from 154 catalogued AGN inside the horizon down to 27 — costs only 1.11× in $\sigma(f_{\rm AGN})$ (§5). But that near-immunity to incompleteness exists only because the completion is handed the true tracer density; when the density is inferred instead, the significance with which an AGN-hosted fraction is detected collapses from 8.2σ to 1.2σ, and the practical requirement turns out to be an external density measurement good to about 10% (§6).

Two further findings are quantitative results in their own right and we report them as such rather than as caveats. First, on a mock intended to match the inference in every ingredient, the recovered expansion rate is low by 1.57 ± 0.184 km s^{-1} Mpc^{-1} — and the budget of that offset now closes (§7). It decomposes into two respects in which the mock was not a draw from the assumed model — a detection rule acting on latent data ($+0.768 \pm 0.226$ km s^{-1} Mpc^{-1}) and a sky-localisation width derived from latent parameters (-0.49 ± 0.08, established with an exact evaluation of the likelihood) — plus an estimator overhead of -0.31 ± 0.13 that is itself budgeted. With the generator repaired, the exact likelihood recovers the truth (-0.06 ± 0.07), and every matched-mock campaign in this paper — the two-tracer measurement, the completeness ladder, the density grids and the realisation ensemble — was regenerated and rerun on the repaired generator. The recoveries those campaigns quote are therefore absolute, with the measured estimator overhead (-0.31 ± 0.13, dominated by the $1/N_{\rm eff}$ selection-variance correction at -0.12 km s^{-1} Mpc^{-1}) as the honest floor beneath them. A companion mechanism on the clustered mocks — a true-redshift detection cut expressed as a distance-space selection — is diagnosed and repaired in §4, where it also yields the one genuinely multitracer lesson of the error budget: the sparse tracer's near-counterpart redshift anchors make the per-event term robust against exactly this class of inconsistency. Second, the host catalog is itself a random realisation, and its realisation contributes 0.97 km s^{-1} Mpc^{-1} of scatter per 1 000-event sample, which credible intervals computed at fixed catalog understate by a factor of 2.2. Any dark-siren error budget that quotes an interval conditional on one catalog is missing a term of that size.

Section 2 sets out the mixture likelihood, the two families of mocks, and the selection integral — including why a sparse tracer needs its own injection proposal, which turns out to change the answer rather than merely the error bar. Sections 3–8 give the results in the order above. Section 9 discusses what the requirements mean for a real AGN survey and what the closure standard the error budget met should look like as a general practice.

2. METHOD

2.1. *A mixture over tracers*

The host of a dark siren is unknown, so the analysis marginalises over it. Given data d_i for event i and parameters Λ containing the cosmology and the source population, the single-event evidence is

$$Z_i(\Lambda) = \int \mathrm{d}\theta\, p(d_i \,|\, \theta)\, p_{\rm pop}(m_1, m_2, \chi \,|\, \Lambda)\, p(z, \hat{n} \,|\, \Lambda), \tag{1}$$

where θ collects source-frame masses, effective spin, redshift and sky direction, and $p(z, \hat{n}|\Lambda)$ is the host prior supplied by the catalog. When more than one tracer of the same underlying structure is available, that prior becomes a mixture,

$$p(z, \hat{n} \,|\, \Lambda) = \sum_{k=1}^{K} f_k\, p_k(z, \hat{n} \,|\, \Lambda), \qquad \sum_k f_k = 1, \tag{2}$$

with f_k the fraction of mergers hosted by tracer k. Throughout we take $K = 2$ with tracer 1 the galaxies and tracer 2 the AGN, so $f_2 \equiv f_{\rm AGN}$ is the parameter of interest and $f_1 = 1 - f_{\rm AGN}$. Written this way, $f_{\rm AGN}$ is a mixture weight between two host priors and nothing else: it does not enter the mass or spin distributions, and it is identified purely by which tracer's redshift-and-sky structure the events prefer.

Each tracer's host prior is built from its pixelated catalog as a kernel-density estimate in redshift, evaluated in the event's HEALPix pixel:

$$p_k(z, \hat{n} \,|\, \Lambda) \;\propto\; g(z; \Lambda) \sum_{j \,\in\, k,\ \hat{n}_j = \hat{n}} \frac{w_j}{W_k} \frac{\mathcal{N}(z; z_j, \sigma_j)}{\zeta_j}, \tag{3}$$

where z_j and σ_j are a catalogued host's redshift and its uncertainty, w_j its weight, $g(z;\Lambda)$ the redshift weighting imposed by the comoving volume element and the merger-rate evolution, and $\zeta_j = \int \mathcal{N}(z; z_j, \sigma_j)\, g(z;\Lambda)\, \mathrm{d}z$ normalises each host's kernel against that weighting.

2.2. *Why the sky normalisation is the whole point*

The one convention in Equation (3) that decides whether a multitracer analysis has any information at all is the choice of W_k. We use the *survey-global* normalisation, $W_k = \sum_j w_j$ over the entire catalog of tracer k, rather than the per-pixel sum. The difference is not cosmetic. With a global normalisation the number of tracer-k hosts in a pixel survives into the likelihood, so a pixel that is rich in galaxies but contains no AGN contributes weight to tracer 1 and none to tracer 2; with a per-pixel normalisation every occupied pixel is renormalised to unity and the sky-density contrast between the tracers is discarded. Since a sparse tracer's information is precisely that most of the sky contains none of it — in our clustered mocks 78.8% of the 49 152 pixels hold no AGN at all, against 0% for the galaxies — the global convention is the one under which $f_{\rm AGN}$ is identifiable. It also means an event localised in a pixel with no catalogued AGN can only be AGN-hosted through the out-of-catalog term of §2.3, which is what couples $f_{\rm AGN}$ to the assumed AGN number density.

2.3. *Missing hosts*

A flux-limited survey stops finding hosts beyond some redshift, and the analysis has to say what is missing. The missing budget per tracer is set by a smooth density model,

$$\frac{\mathrm{d}N_{{\rm miss},k}}{\mathrm{d}z} = n_{0,k}\, \frac{\mathrm{d}V_c}{\mathrm{d}z}\, (1+z)^{\delta_k} - \frac{\mathrm{d}N_{{\rm obs},k}}{\mathrm{d}z}, \tag{4}$$

and distributed isotropically over the sky, so completeness is never a free function of redshift: it follows from the assumed density amplitude and evolution together with the observed counts,

$$C_k(z) = \frac{\mathrm{d}N_{{\rm obs},k}/\mathrm{d}z}{n_{0,k}\, (\mathrm{d}V_c/\mathrm{d}z)\, (1+z)^{\delta_k}}. \tag{5}$$

This is the mechanism that lets a thinned survey still constrain $f_{\rm AGN}$: as observed hosts disappear, each tracer's prior migrates from its own KDE onto its own smooth field, and the two fields still differ by the number-density ratio that carried the signal. It is also the mechanism that makes the answer depend on $n_{0,k}$, which §6 is about. Setting $n_{0,k} \to 0$ removes the missing budget and recovers the complete-catalog estimator, which we use for the complete-catalog results.

Anchoring $n_{0,k}$ fixes the budget's amplitude but not its shape: the true tracer density is whatever the mock's density field produced, not exactly $(1+z)^\delta\, \mathrm{d}V_c/\mathrm{d}z$. We therefore anchor $n_{0,k}$ and δ_k at the best fit of the model *form* to the true host density rather than at the raw mean density — for the single-tracer mock, $\log_{10} n_0 = -5.8064$ and $\delta = +0.019$, against a raw mean density of -5.7995 — and quote the residual of that fit as the level below which an apparent completeness bias is not attributable to completeness. That residual is 2.6% rms over the fit range and 7.2% within the detection horizon, the latter dominated by the shot noise of the 9 105 hosts that lie there. For the two-tracer mock the corresponding anchors are $(\log_{10} n_0, \delta) = (-5.80638, +0.0194)$ for the galaxies and $(-7.72003, -0.0031)$ for the AGN, with shape residuals of 6.7% and 8.3% inside the horizon.

2.4. *Selection*

With detection accounted for, the likelihood over $N_{\rm obs}$ detected events is

$$\ln \mathcal{L}(\Lambda) = \sum_{i=1}^{N_{\rm obs}} \ln Z_i(\Lambda) - N_{\rm obs} \ln \mu(\Lambda), \qquad (6)$$

with $\mu(\Lambda) = \int \mathrm{d}\theta \, P_{\rm det}(\theta)\, p(\theta|\Lambda)$ the detectable fraction (Mandel et al. 2019). We estimate μ by importance sampling a campaign of injections recorded in the cosmology-free observables $(m_1^{\rm det}, q, d_L)$ with their proposal density, and require the estimator's effective sample size to satisfy $N_{\rm eff} > 5\, N_{\rm obs}$ for the estimate to be usable (Farr 2019; Vitale et al. 2022). Grid cells that fail that criterion are inadmissible and are shown as such in the figures; where they occur we report how much posterior mass lies next to them, because a posterior whose peak sits against an inadmissible edge is a statement about the selection estimate rather than about the data.

2.5. *A sparse tracer needs its own injections*

The selection integral in Equation (6) is conditioned on the host catalog through Equation (3), so an injection carries weight only if its redshift lands within a few σ_j of an actual catalogued host *in its own pixel*. As $f_{\rm AGN}$ rises, μ leans on the sparse tracer, and injections drawn from the source population essentially never land on it. This is not a nuisance: it changes the answer. We therefore draw injections from a three-branch mixture — 0.65 from the source population, 0.10 uniform, and 0.25 placed at catalogued AGN — and store, for every injection, the exact mixture density at its own coordinates. Two properties make the construction safe. The targeted branch reads the *pixelated survey file*, which is the object the likelihood conditions on and whose stored redshifts and widths are the kernel centres and widths of Equation (3), so proposal and target cannot drift apart through a pixelisation or kernel-width mismatch; and the stored densities were recomputed by an independent code path, agreeing to a maximum relative error of 5.5×10^{-14}. From 120 million proposals, 351 919 are detected, of which 20.1% fall on AGN catalog support, against 1.0% for population draws and 2.6% for uniform ones, while the detected fraction 2.9×10^{-3} stays within a few per cent of the untargeted campaign's.

Across a completeness ladder the support the likelihood conditions on shrinks with the survey, so one injection set per rung is generated against that rung's own catalogs. Holding the branch *weights* fixed along the ladder is the wrong invariant: a flux limit leaves only bright, nearby hosts, so the targeted branch's detection efficiency climbs steeply and, at fixed weights, the campaign fills with targeted rows carrying negligible weight while starving the population branch — the one branch that covers the out-of-catalog term, which is exactly the term that grows as the survey empties. We hold the fraction of *detected* rows contributed by each branch fixed instead, solving for the weights from a per-rung pilot run.

2.6. *The catalog's redshift resolution is not free*

The kernel widths σ_j in Equation (3) have to sit inside a window bounded on both sides, and both bounds are measured.

The lower bound is computational, and it is a genuine design tension rather than a numerical detail. Narrow kernels — spectroscopic redshift errors, say — make each tracer's prior a set of thin shells in redshift, which sharpens the dark-siren information but shrinks the support the selection integral has to cover, driving $N_{\rm eff}$ down and the per-event Monte Carlo variance up. On the matched single-tracer mock, pixelating the same catalog with its survey-style spectroscopic widths (median 2.5×10^{-3}) instead of our adopted $\mathrm{d}z = 3.0 \times 10^{-3}$ cuts the selection integral's effective sample size at the truth from 9 956 to 2 295 — below the $5\, N_{\rm obs} = 5\,000$ floor, so the scan is not even admissible — while quadrupling the summed per-event Monte Carlo variance (12 to 50). Widening the kernels buys that resolution back at the direct cost of the signal, because the catalog's redshift precision *is* the dark-siren information. At the adopted width the two-tracer complete rung has $N_{\rm eff} = 147 \times 10^3$ against a floor of 1000, with a summed per-event log-likelihood variance of 6.1.

The upper bound is a threshold in the inference itself, not a slow degradation, and it is set by the scale the kernels must never approach: the parameter estimates' own redshift resolution, $\sigma_z^{\rm PE} \simeq \sigma_{d_L}\, d_L\, (\mathrm{d}d_L/\mathrm{d}z)^{-1}$, which is 0.012 at the median event of the matched mock. Broadening the catalog kernels at fixed data leaves the recovered expansion rate flat within 0.22 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ up to an effective kernel width of 0.020 — still under the PE resolution — and then fails catastrophically: -2.22 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 0.030, -4.73 at 0.040, -9.29 at 0.070, reproduced on a second realisation (-4.16 at 0.040; Figure 1). Once the host prior is as broad as the measurement, the likelihood's redshift information comes from the prior's overlap with the moving posterior rather than from the hosts, and the distance scale runs away. The design rule for a real survey follows di-

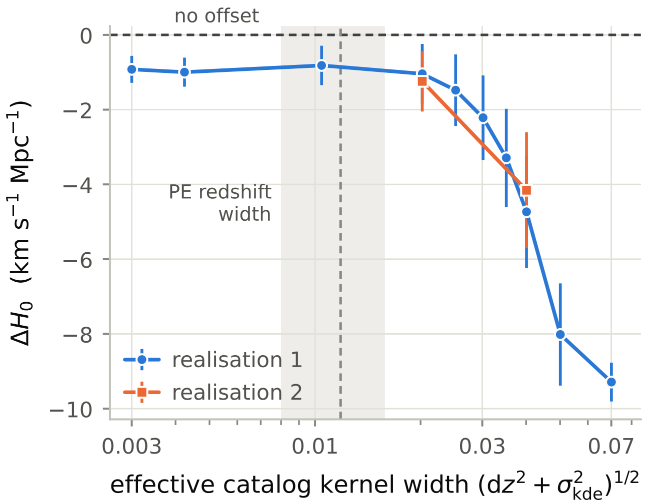

Figure 1. The catalog kernel width is a threshold lever. Recovered expansion-rate offset against the effective kernel width $(\mathrm{d}z^2+\sigma^2_{\mathrm{kde}})^{1/2}$ when the catalog kernels are broadened at fixed data, for two independent realisations of the matched single-tracer mock. The band marks the 16–84% range of the per-event PE redshift width $\sigma_{d_L}\, d_L\,(\mathrm{d}d_L/\mathrm{d}z)^{-1}$; the offset is flat until the kernels approach it, then runs away.

rectly, and it is a statement about photometric redshifts: catalog redshift kernels — including any photo-z scatter they encode — must stay well below the distance posteriors' implied redshift width, or the catalog stops being a redshift prior and becomes a redshift bias.

2.7. *Mocks*

Two families of mocks are used, and they answer different questions.

Clustered two-tracer mocks. The flagship $K = 2$ measurements use lognormal density fields generated with GLASS (Tessore et al. 2023), sampled into two biased tracers: 1 177 289 galaxies with linear bias 1.2 and 11 724 AGN with linear bias 2.0, out to $z = 1.5$, on a HEALPix grid with $N_{\mathrm{side}} = 64$. The resulting comoving number densities are $\log_{10} n_0 = -5.506$ and -7.508, a ratio of 100, and 78.8% of pixels contain no AGN. Events are 1 000 mergers whose hosts are drawn from the two host lists in the planted proportion, with source redshifts restricted to $z \leq 1.0$; masses and spins are drawn from a power-law-plus-peak population at fiducial values, and each event carries 2 000 posterior samples built by exact inverse-transform sampling of the flat-prior posterior given a noisy observation, with a fractional distance uncertainty of 0.10. These mocks carry the clustering-bias contrast, which is the physical channel the paper is about; their detection rule is a hard cut on true source redshift, which is the one place where their generative process is not the inference's model. Section 4.2 measures exactly what that inconsistency does to the recovered distance scale, and §9.5 states what the clustered mocks can and cannot be used for because of it.

Matched mocks. To measure the error budget of the inference itself we need mocks whose generative process *is* the assumed model, ingredient by ingredient. These are generated end to end by the inference code's own mock generator: detection is a noisy network signal-to-noise threshold on an observable, hosts are accepted with the same redshift weighting the likelihood assumes, and masses, spins and posterior samples come from the same samplers the likelihood uses. That standard is not automatic even for a generator built for it: §7 shows the generator as first shipped violated it in three measured respects — posterior samples centred on the truth, a detection decision taken on latent data, and a sky-localisation width derived from latent parameters — each identified and repaired there. Except where a comparison against the original generator is itself the point, every matched-mock result in this paper comes from the repaired generator, in which every quantity the likelihood conditions on is a function of the observed data. The single-tracer version has 1 000 000 hosts out to $z = 2.0$ — far beyond the events, so the catalog's redshift boundary cannot shape the prior where the data live — pixelated complete at $N_{\mathrm{side}} = 16$ with 0% empty pixels, 1 000 events with 2 000 posterior samples each and a fractional distance uncertainty of 0.10, and a selection campaign of 120 million proposals of which 343 702 are detected. The two-tracer version has 1 000 000 galaxies and a nested subset of 12 000 AGN ($\log_{10} n_0 = -5.80$ and -7.72, with 37.5% of AGN pixels empty), 140 galaxy-hosted plus 60 AGN-hosted events at a planted $f_{\mathrm{AGN}} = 0.300$, and $N_{\mathrm{side}} = 32$. Their scope limit is stated up front: the generator's catalogs are unclustered, so in these mocks f_{AGN} is identified by the number-density contrast alone with none of the bias contrast the clustered mocks carry. The two families are complementary, not redundant.

The completeness ladder. Incompleteness is imposed as an isotropic apparent-magnitude limit on the complete catalog — no footprint, no hard redshift cut — so that completeness is a *consequence* of survey depth and has the declining shape a flux-limited survey actually has. The events are untouched: incompleteness is an observational effect, so the mergers are exactly the ones that happened, and only the catalogs the inference sees are thinned. In the two-tracer ladder the AGN are a nested subset of the galaxies and inherit their host galaxy's apparent magnitude, so one flux limit thins both tracers with the same $C(z)$ and the ladder stays a single-axis experiment.

2.8. *Inference*

We evaluate the likelihood of Equation (6) directly on grids rather than sampling it, which is affordable because at most three parameters are free at a time and which removes any question of sampler convergence from the comparisons. Flat priors are used throughout, and we quote medians with equal-tailed 68% and 90% intervals; where the truth lies on a prior boundary we say so and quote the one-sided limit instead. $\Omega_{m,0}$ is pinned at 0.3075 and the mass and spin population is held at the mock's truth, so the mass distribution is informative but not free; the fiducial expansion rate is $H_0 = 67.74\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. Unless stated otherwise the sampled parameters are H_0 and f_{AGN}, with the tracer densities anchored as in §2.3; §6 adds $\log_{10} n_{0,\mathrm{AGN}}$.

3. THE AGN-HOSTED FRACTION IS MEASURABLE ACROSS ITS WHOLE RANGE

On complete clustered catalogs, 1 000 dark sirens recover the AGN-hosted fraction everywhere between almost none and all of the population. Scanning f_{AGN} at the true expansion rate over 41 points gives medians of 0.0195, 0.3221, 0.6872 and 0.9750 against planted values of 0.00989, 0.307, 0.703 and 1 (Table 1, Figure 2). The planted value lies inside the 68% interval at every interior point; the $f_{\mathrm{AGN}} = 1$ case sits on the prior boundary, where an equal-tailed interval cannot cover it and the one-sided 68% limit, 0.966, does. The residual offsets are +0.010, +0.015, −0.016 and −0.025, none of them larger than 0.025 and all comparable to the 68% half-widths, which run from 0.016 to 0.025.

What makes this work is the contrast between the two host priors and not the sparse tracer's redshifts on their own. The AGN catalog contains 11 724 objects against 1 177 289 galaxies and leaves 78.8% of the sky without a single catalogued AGN, so on its own it localises almost nothing; what it does do is make a large part of the sky implausible for an AGN-hosted event while the galaxy catalog covers all of it. The mixture weight is the parameter that measures how often events prefer the sparse tracer's structure, and at these densities it is measured to 0.016–0.025 from 1 000 events.

Table 1. The AGN-hosted fraction recovered from the clustered two-tracer mock at the true expansion rate, on complete catalogs.

planted f_{AGN}	median	68% interval	offset
0.0099	0.0195	[0.006, 0.039]	+0.010
0.3070	0.3221	[0.300, 0.346]	+0.015
0.7030	0.6872	[0.662, 0.712]	−0.016
1.0000	0.9750	[0.966, 1]†	−0.025

NOTE—† At $f_{\mathrm{AGN}} = 1$ the truth lies on the prior boundary, where an equal-tailed interval cannot cover it; the one-sided 68% lower limit is quoted instead. Each row is a 41-point scan of 1 000 events.

3.1. *The joint plane, and a distance-scale deficit*

Freeing the expansion rate together with the fraction leaves the fraction essentially unchanged and exposes a systematic in the distance scale (Figure 3). At a planted $f_{\mathrm{AGN}} = 0.307$ the joint grid gives $H_0 = 66.81^{+0.27}_{-0.26}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $f_{\mathrm{AGN}} = 0.325 \pm 0.020$ with a correlation coefficient of $\rho = -0.01$; at $f_{\mathrm{AGN}} = 0.703$ it gives $H_0 = 64.36^{+0.28}_{-0.23}\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, $f_{\mathrm{AGN}} = 0.711 \pm 0.018$ and $\rho = -0.30$. The near-zero correlation at the lower fraction is worth stating on its own: there, the mixture weight is measured essentially independently of the distance scale, so the two parameters are not competing for the same information.

The expansion rate, however, is recovered low — by 0.93 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $f_{\mathrm{AGN}} = 0.307$ and 3.38 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $f_{\mathrm{AGN}} = 0.703$, which is 3.5 and 13.4 times the respective 68% half-widths — and the deficit grows as the mixture leans on the sparse tracer. It is not a property of the mixture weight: f_{AGN} is recovered correctly both at fixed H_0 and along the joint surface, and $\rho \simeq 0$ at the lower fraction. Two explanations that suggest themselves are measurably wrong. The host redshift weighting is one: the events' hosts were drawn uniformly over eligible catalog entries, corresponding to a rate index one unit away from the value the likelihood assumes, and repeating the scans at the matching value moves H_0 by at most 0.16 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, because under the global sky normalisation the factor enters both the per-event host prior and the survey-wide normalisation and cancels. The catalog's redshift boundary is the other, and it is not innocent: truncating both catalogs at $z \leq 1$ with the events held fixed moves H_0 by −4.09 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $f_{\mathrm{AGN}} = 0.307$. A 4.1 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ shift from relocating a catalog-construction boundary, with the data unchanged, says the inference is sensitive to how the host prior is terminated well outside the range the events occupy; §4.2 explains why.

Splitting the likelihood at fixed mixture weight into its per-event and selection terms localises where that sensitivity lives. At $f_{\mathrm{AGN}} = 0.307$ the per-event term alone peaks at 70.83 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and the selection term shifts the total by −4.01 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, landing at 66.82; at $f_{\mathrm{AGN}} = 0.703$ the same numbers are

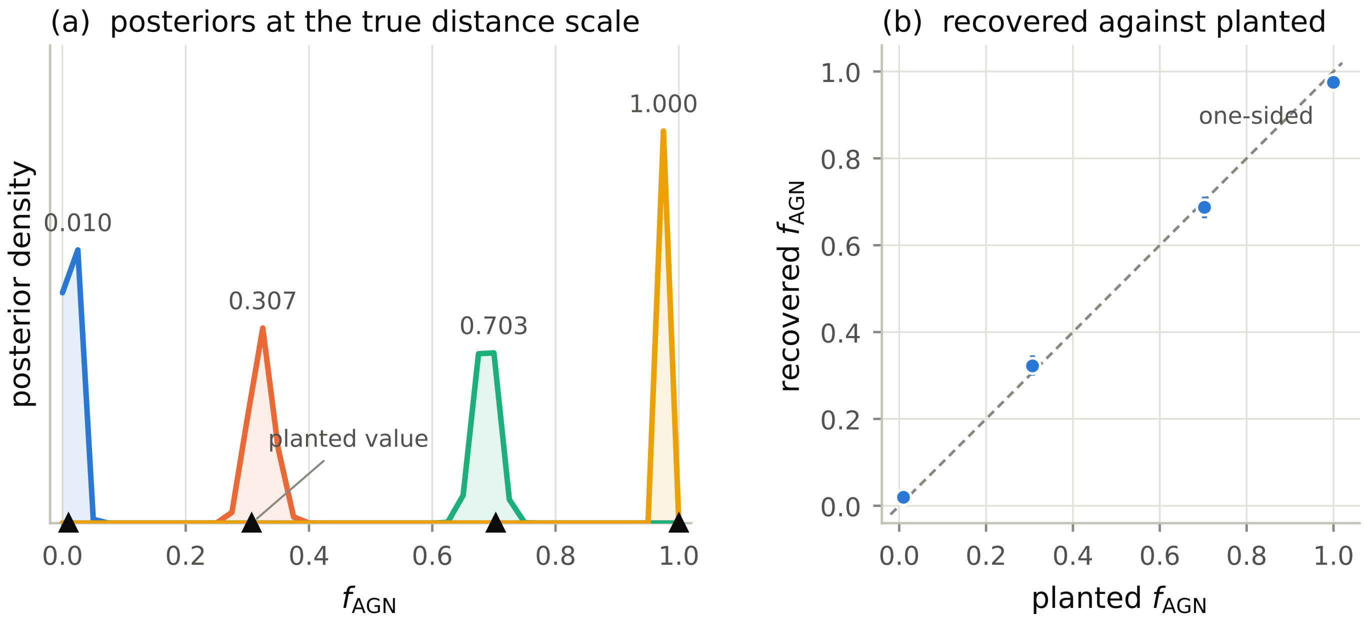

Figure 2. Recovery of the AGN-hosted fraction from 1 000 dark sirens on complete clustered catalogs. *(a)* the flat-prior posteriors from 41-point scans at the true expansion rate, one per planted fraction (triangles). *(b)* recovered against planted, with 68% intervals; the $f_{\rm AGN} = 1$ point lies on the prior boundary and carries its one-sided limit. No likelihood evaluation is inadmissible in any of these scans.

68.69, -4.27 and 64.42. The selection term supplies $\sim$ 4.01 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ of pull almost independently of the fraction, with $\ln\mu$ spanning 0.36 nats over the scanned range against $N_{\rm obs} = 1\,000$, a lever of $\sim$ 356 nats; the per-event term carries all of the f-dependence. This localises the sensitivity but does not by itself identify an error, because the two terms are meant to oppose one another and only their sum has to be unbiased. The accurate-looking result at $f_{\rm AGN} = 0.307$ is accurate by cancellation, not by correctness: correcting only the selection normalisation would move it from -0.93 to $+3.09$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Section 4.2 resolves both terms: the two pulls are spurious levers sprung by a single inconsistency between the mocks' true-redshift detection cut and the likelihood's distance-space selection and host prior, and an estimator matched to the detection rule recovers the truth at both planted fractions. The f-dependence visible above is the collapse of the numerator's pull as the mixture weight moves onto the sparse tracer (§4.3). This family of offsets is specific to the clustered mocks' detection rule; the matched mocks of §4–8 carry no analogue of it, their own error budget is measured and closed in §7, and the absolute recoveries they quote sit on the measured estimator-overhead floor (-0.31 ± 0.13 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$) rather than on an unexplained offset.

4. THE SELECTION INTEGRAL DECIDES THE ANSWER

Three results in this paper are, at bottom, statements about the selection integral. The first is practical: on a sparse tracer the selection Monte Carlo must follow the catalog or the measurement is invalid (§4.1). The second is a mechanism: the clustered mocks' distance-scale deficit of §3.1 is the incomplete cancellation of two spurious levers sprung by a single inconsistency between the detection rule and the likelihood's distance-space selection (§4.2). The third is the one genuinely multitracer effect of the three: the sparse tracer, precisely because it is sparse, anchors the distance scale against that inconsistency, and the growth of the deficit with $f_{\rm AGN}$ is the loss of an accidental cancellation, not a new pathology (§4.3).

4.1. *A sparse tracer needs its own proposal*

On the matched two-tracer mock, with 200 events at a planted $f_{\rm AGN} = 0.300$, injections drawn from the source population give an effective sample size for the selection integral that *falls* as the mixture leans on the sparse tracer: 2696 at $f_{\rm AGN} = 0$, then 497, 92 and 41 at $f_{\rm AGN} =$ 0.3, 0.7 and 1, against a floor of 1000 — a decay by a factor of 66 across the grid, with three of the four points below the floor (Figure 4a). Adding a branch that places 0.25 of the proposal at catalogued AGN reverses the trend: 2170, 4977, 25198 and 65509, a *rise* by a factor of 30. The honest cost sits at $f_{\rm AGN} = 0$, where moving proposal weight off the dense tracer gives up 20% of its

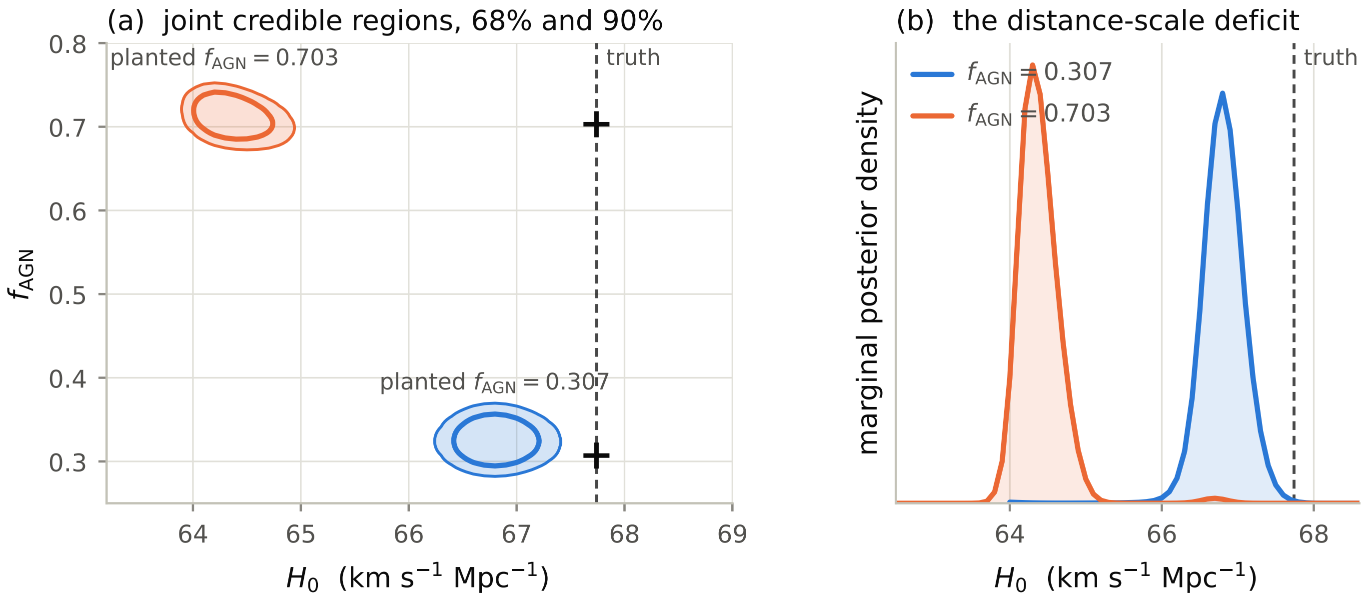

Figure 3. Joint constraint on the expansion rate and the AGN-hosted fraction from 1 000 dark sirens on complete clustered catalogs, at two planted fractions. *(a)* 68% and 90% credible regions on shared axes; crosses mark the truth. *(b)* the marginal distance-scale posteriors. The fraction is recovered at both values while the expansion rate is low by 0.93 and 3.38 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ respectively, the deficit growing as the mixture leans on the sparse tracer.

effective sample size and still clears the floor by a factor of 2.2.

What that buys is not precision but validity. At the reduced sample of 80 events, where the untargeted campaign is still usable at all, only 14 of 41 grid cells are admissible, the admissible range ends at $f_{\rm AGN} = 0.325$, and the posterior peaks against that very edge: it reports $f_{\rm AGN} = 0.3152$ with a 68% interval of $[0.289, 0.338]$, which contains the planted value and means nothing, because the interval is set by where the selection estimate stopped being usable (Figure 4b). With the same events and only the injection set changed, every cell becomes admissible and the peak moves from $f_{\rm AGN} = 0.3152$ to 0.2157 — about two half-widths, so the campaign changes the answer and not merely its reliability. The measurement itself is made on the repaired-generator events of §7.4, whose detected set, catalogs and injection campaigns are identical to the exhibit above and whose posterior construction is what the repair fixed: the targeted campaign gives $f_{\rm AGN} = 0.2707$ $[0.218, 0.326]$ at 80 events, and at the full sample — which the targeted campaign is what makes reachable — $f_{\rm AGN} = 0.2629$ with $[0.228, 0.299]$: 0.037 below the planted value on a 0.036 half-width, 1.0σ, with the planted value inside the 90% interval.

The joint plane on this mock is interpretable for the first time with the targeted campaign: $H_0 = 68.31$ $[67.72, 68.92]$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $f_{\rm AGN} = 0.2652 \pm 0.036$ with $\rho = +0.07$. Of 3321 cells, 1478 are inadmissible, but all of them lie at expansion rates far from the peak: the posterior mass adjacent to the inadmissible region is 7.9×10^{-9}, and none of the 68% region touches it. Both parameters land within about one half-width of the truth — $+0.57$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ on the expansion rate (1.0σ) and 1.0σ on the fraction — and with $\rho \simeq 0$ they are measured essentially independently. The same scans on the mock as first generated had returned H_0 1.34 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ low (2.7σ) and the fraction 1.8σ low; the sky-width repair of §7.4, which changes only the fidelity of the mock's posterior construction, removes both, so the two-tracer measurement inherits the closure of the single-tracer budget rather than merely being consistent with it.

Two points are worth carrying forward. First, this is the second place in this programme where the binding constraint on a field-mode multitracer measurement was the resolution of the selection Monte Carlo rather than the gravitational-wave data; it appears to be a structural property of conditioning the selection integral on a sparse catalog, and any analysis that does so needs a proposal that follows the tracer. Second, the complete-catalog limit obtained by driving the assumed density to zero is far worse conditioned than the incomplete model evaluated at a true density anchor: on the same mock the anchored incomplete model reaches $N_{\rm eff} = 147 \times 10^3$ against 4977 for the complete-catalog trick. The trick buys an unbiased complete-catalog limit at a real cost in selection resolution.

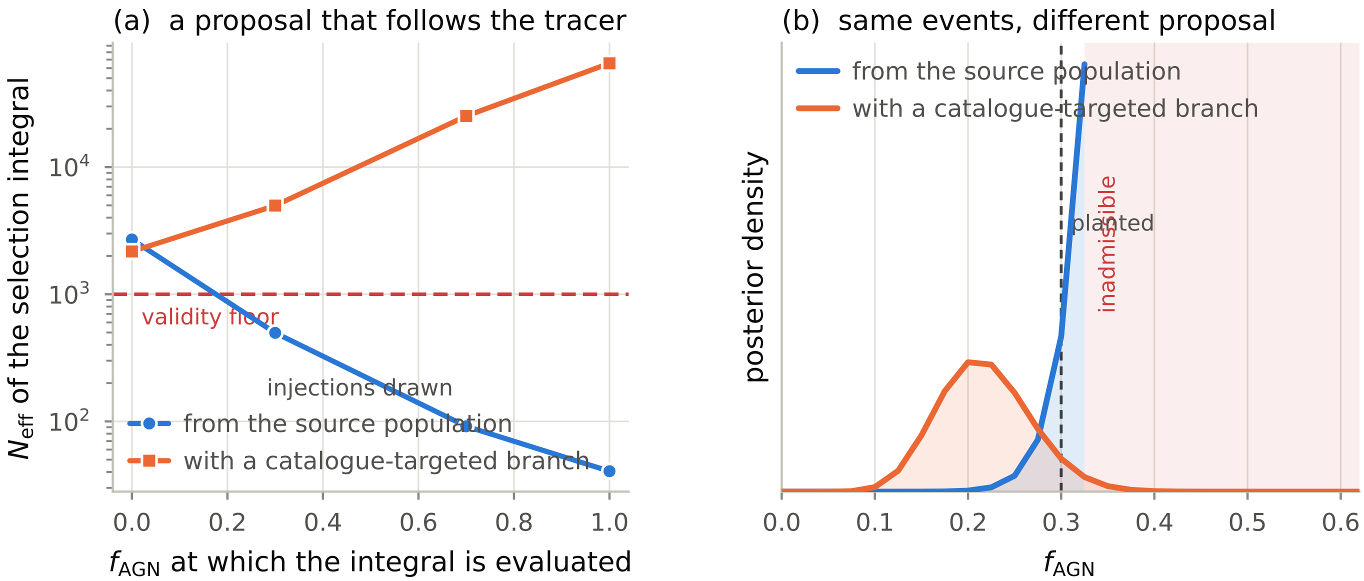

Figure 4. Why a sparse tracer needs its own selection proposal, on the matched two-tracer mock. *(a)* effective sample size of the catalog-conditioned selection integral against the fraction at which it is evaluated, at the full sample of 200 events, for injections drawn from the source population and for a proposal that places 0.25 of its draws at catalogued AGN; the dashed line is the reliability floor $5\,N_{\rm obs}$. *(b)* the fraction's posterior from the same 41-point scan with the two campaigns, at the reduced sample where the untargeted one is still evaluable at all; shading marks inadmissible cells. The untargeted posterior peaks against the edge of the admissible region, so its apparent agreement with the planted value carries no information. The comparison holds the events fixed (the mock as first generated); the generator repair of §7.4 changes neither the detected set nor the injections, so it leaves this campaign-versus-campaign statement untouched.

4.2. *One inconsistency, two levers: the clustered mocks' deficit*

The distance-scale deficit of §3.1 — H_0 low by $0.93\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $f_{\rm AGN} = 0.307$ and 3.38 at 0.703, with the fraction recovered correctly — has a complete mechanical explanation, obtained from a term-by-term rebuild of the likelihood that reproduces the production curves to better than 10^{-3} nats and can then be evaluated with counterfactual switches. The origin is one inconsistency: the mocks detect events by a cut on *true source redshift*, $z \le 1.0$, while the likelihood expresses both its selection integral and its host prior in *luminosity distance*, against catalogs whose support continues to $z = 1.5$ — 57% of the galaxies and 57% of the AGN lie above the horizon no detected event can occupy. That single mismatch springs two opposing levers.

The selection lever (-4.01 / $-4.27\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ *at the two planted fractions).* Because the injections carry exact distances at the true cosmology, the detected injection set fills exactly $d_L \le 6\,798$ Mpc, the distance of the horizon. Under the analysis model at trial H_0 that fixed distance boundary sits at a redshift $z^*(H_0)$ that sweeps from 0.91 at $H_0 = 60$ to 1.09 at 75: raising H_0 pushes z^* up the still-rising catalog $\mathrm{d}N/\mathrm{d}z$, so the detectable fraction μ *rises* with H_0 ($\mathrm{d}\ln\mu/\mathrm{d}H_0 = 0.0245$ at truth) when for a true-redshift cut it should be flat. Multiplied by $-N_{\rm obs}$ this is a -24 nats per $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ lever. A zero-parameter analytic model — the catalog's redshift distribution integrated to $z^*(H_0)$ with the population's rate weighting — predicts a slope of 0.0234, 96% of the measured value, and re-peaking the likelihood against it reproduces -3.89 of the -4.01 shift. The lever is not an artifact of injection coverage: recomputing $\mu(H_0)$ with an untargeted, independently seeded injection set changes the slope by 0.2%. The estimator faithfully computes the model's μ; the model's μ is what is wrong for this detection rule.

The numerator lever ($+3.72$ / $+1.19\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, *opposing).* Events exist only below the horizon, but the host prior does not: the broad distance posteriors of near-horizon events map part of their mass to $z(d_L; H_0) > 1$, where the catalogs still offer (spurious, for detected events) support. The mean posterior mass beyond the horizon is 8.1% at truth and 17.8% at $H_0 = 75$ (at $f_{\rm AGN} = 0.307$; 5.4% and 13.2% at 0.703), so raising H_0 maps more posterior mass higher up the rising $\mathrm{d}N/\mathrm{d}z$ and the per-event term grows with H_0. Masking the host prior above the horizon removes essentially the whole numerator offset ($+3.09 \to -0.63$). This also explains the catalog-edge lever of §3.1: truncating the catalogs at $z \le 1$ removed the numerator's upward pull while leaving the selection slope uncompensated, which is why the data-free relocation of a catalog boundary moved H_0 by $-4.09\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The two levers account for the deficit in full. Repairing both at once — a host prior truncated at the detection horizon together with the H_0-independent selection normalisation the true-redshift cut actually implies — returns the peak to $-0.63 \pm 0.22\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ of the truth at $f_{\rm AGN} = 0.307$ and -0.24 at 0.703 (Figure 5); repairing either alone makes the bias worse, because the levers are opposed. The delta-method Monte Carlo bias of the per-event estimates contributes at most $0.07\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. For real data the transferable statement is about consistency rather than about this mock: the selection function must be expressed in the variable that actually decided detection, and a host prior with support well beyond the detected population's horizon hands the numerator spurious H_0 information of order the posterior-mass leak whenever the distance posteriors are broad.

4.3. *The sparse tracer anchors the distance scale*

Why does the deficit grow with $f_{\rm AGN}$ when both levers are set by the same catalogs? Not through the redshift distributions — the two tracers' $\mathrm{d}N/\mathrm{d}z$ shapes are identical to within a fraction of a per cent (57% versus 57% of objects above the horizon) — and indeed the selection lever barely moves ($-4.01 \rightarrow -4.27$). What changes is the numerator, and the reason is sparseness. With 11 724 AGN spread over 49 152 pixels, 78.8% of the sky holds none, and within an event's localisation the AGN branch of the mixture is a handful of near-delta-function redshift kernels of width $\sigma_z = 3.0 \times 10^{-3}(1+z)$: effectively a noisy counterpart prior. It pins the distance scale. The per-event term of AGN-hosted events peaks at $68.79\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (+1.05 from truth), while that of galaxy-hosted events — whose smooth, deep prior is maximally exposed to the above-horizon support — peaks at 79.0 (+11.3). As the planted fraction rises from 0.307 to 0.703 the mean posterior weight on the anchored branch grows from 0.32 to 0.69, the numerator's spurious pull collapses from +3.09 to $+0.95\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ and its curvature nearly doubles ($10.1 \rightarrow 18.4$ nats per $(\mathrm{km\,s^{-1}\,Mpc^{-1}})^2$), while the selection lever stays put: the net deficit deepens from -0.93 to -3.38. The value at $f_{\rm AGN} = 0.307$ is accurate by cancellation, not by correctness. The offset at the higher fraction is robust to the selection realisation: three independent injection sets give $-3.2 \pm 0.3\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.

Read as multitracer physics rather than as a bug report, this is a positive result. A sparse, strongly clustered tracer contributes exactly what a statistical analysis lacks: near-counterpart redshift anchors that make the per-event term robust against mis-modelled prior support. The anchoring is invisible in a correct analysis — the repaired estimator recovers the truth at both fractions — but it means that the population most often analysed, dense smooth catalogs at $f_{\rm AGN} = 0$, is also the configuration most exposed to selection–prior inconsistencies of this kind.

5. WHAT INCOMPLETENESS COSTS

5.1. *One tracer: the missing-host model does not add bias*

The first question is whether the completion of §2.3 works at all when it is handed the correct missing-host budget. Imposing an isotropic flux limit on the matched single-tracer catalog takes the completeness within the detection horizon from 100% down to 21.9%, with the surviving host count falling from 1 000 000 to 2 236 and the fraction of sky pixels containing no catalogued host rising from 0% to 47.4% (Figure 6a, Table 2). Over that range the recovered expansion rate stays within 0.5σ of the complete-catalog control, the largest departure being $0.77\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Figure 6b), and no likelihood evaluation is inadmissible at any level. Replacing three-quarters of the observed hosts inside the horizon by a modelled budget, and leaving nearly half the sky without a catalogued host, adds no detectable bias to the distance scale.

The price is width. The 68% interval grows by up to $2.3\times$ (Figure 6c), which is what one expects when observed hosts are replaced by a smooth prior, and it does not grow monotonically: the per-level precision is a single realisation, with half-widths between 0.65 and $1.52\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, so the level-to-level ordering of the widths is not itself meaningful.

Two limits on this statement. Every level, including the complete-catalog control, sits low by roughly 0.78–$1.55\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$; that pedestal is the offset of §7 and not a completeness effect, which is why the comparison here is differential. And with per-level precision of order 0.65–$1.52\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$, a bias smaller than about $0.65\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ would not have been detected; the density model's own shape residual, 7.2% within the horizon, sets a second floor of the same character.

5.2. *Two tracers: the fraction barely notices*

The measurement that matters for survey design is what incompleteness does to $f_{\rm AGN}$, and the answer is much less than what it does to H_0. Putting the matched two-tracer mock through the same flux-limit ladder — the same events throughout, only the catalogs thinned

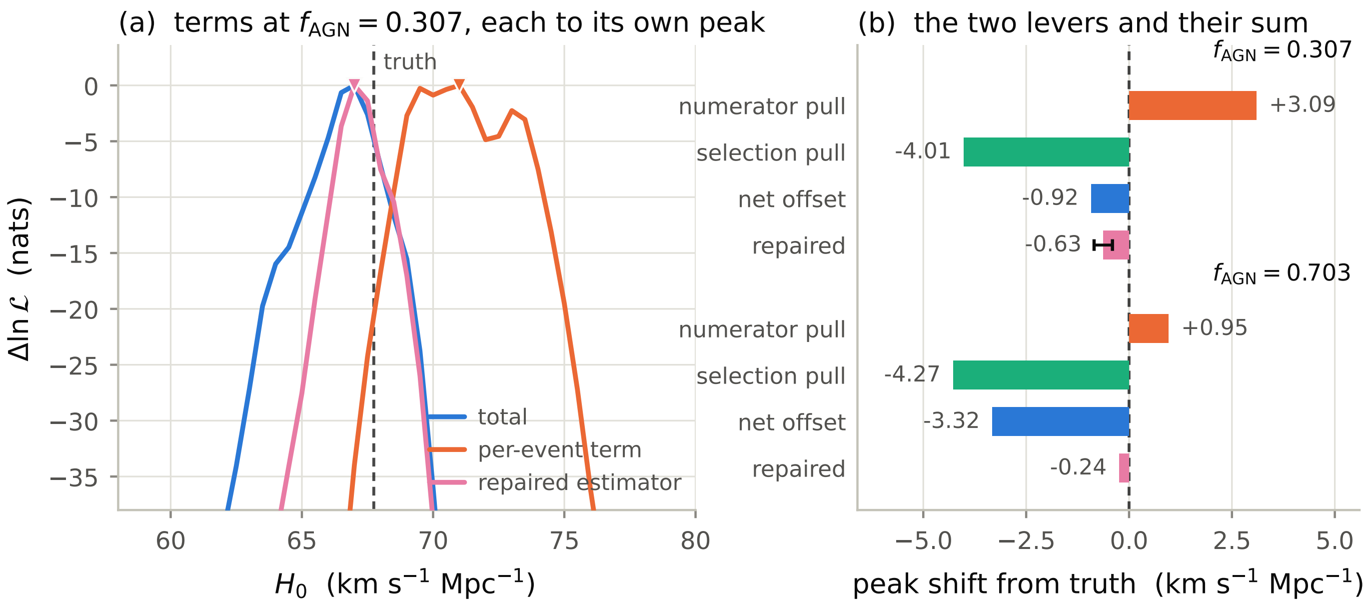

Figure 5. The clustered mocks' distance-scale deficit decomposed, at fixed planted fractions. *(a)* log-likelihood terms against H_0 at $f_{\rm AGN} = 0.307$, each normalised to its own peak: the per-event term peaks $+3.09$ km s^{-1} Mpc^{-1} high, and the selection term — a near-linear tilt of ~ 24 nats per km s^{-1} Mpc^{-1} with no interior peak, visible here as the displacement between the per-event term and the total — pulls the total down by -4.01. The repaired estimator — host prior truncated at the detection horizon, selection normalisation matched to the true-redshift cut — peaks at the truth. *(b)* the same budget at both planted fractions: the selection lever is nearly independent of the fraction while the numerator pull collapses as the mixture weight moves onto the sparse tracer (§4.3), so the net deficit grows.

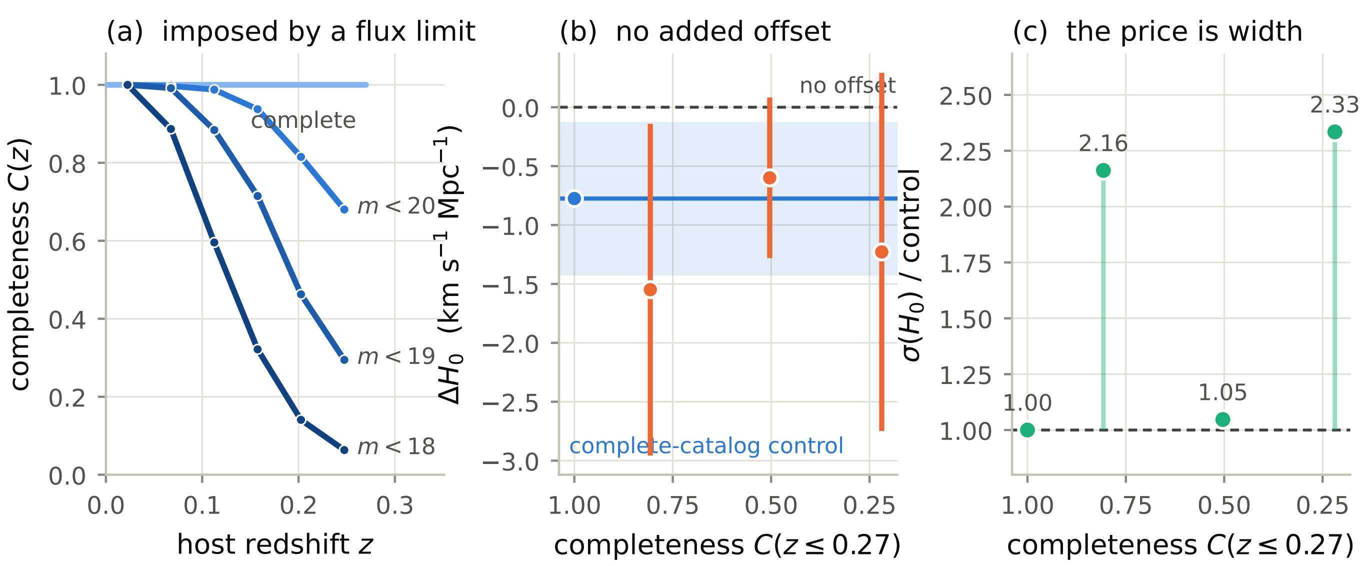

Figure 6. Single-tracer flux-limit ladder with the tracer density anchored at the truth. *(a)* the completeness the flux limits impose across the detection horizon, which declines with redshift as a flux-limited survey's does rather than being cut off. *(b)* the recovered expansion rate at each level; the band is the complete-catalog control's own 68% interval, and the largest departure from it is 0.5σ. *(c)* the 68% half-width as a factor against that control. Error bars and the band are 68% intervals from single realisations, so the ordering of the widths between levels is not significant; the range, up to $2.3\times$, is.

Table 2. Single-tracer flux-limit ladder. Completeness C is quoted within the detection horizon $z \leq 0.27$. Offsets are relative to the true expansion rate; the last three columns are differential against the complete-catalog control.

level	hosts	C (%)	empty sky (%)	ΔH_0	vs. control	σ	width $\times$
complete	1 000 000	100.0	0.0	-0.78 ± 0.65	$+0.00$	0.0	1.00
$m < 20$	20 369	80.7	0.0	-1.55 ± 1.41	-0.77	0.5	2.16
$m < 19$	7 019	50.4	9.8	-0.60 ± 0.68	$+0.18$	0.2	1.05
$m < 18$	2 236	21.9	47.4	-1.23 ± 1.52	-0.45	0.3	2.33

— takes the completeness within the range the events occupy from 1.000 to 0.175, the catalogued AGN inside the horizon from 154 to 27, and the fraction of sky with no catalogued AGN from 37.5% to 99.8%. Across that 5.7-fold loss of sparse-tracer hosts, the joint 68% half-width on the fraction grows by 1.11× while that on the expansion rate grows by 1.87× (Figure 7a, Table 3).

Three features of that ladder are worth stating plainly.

The first rungs are nearly free for the distance scale. $\sigma(H_0)$ grows by only 1.08× at the first rung and reaches 1.87× only at the faint end, non-monotonically in between; as in §5.1 the per-level widths are single realisations, so the rung-to-rung ordering is not itself meaningful, while the overall range is. The reason the cost is so low is that the hosts a flux limit removes first lie beyond the gravitational-wave horizon, where they contribute prior weight at redshifts no event can occupy; only once the limit starts removing hosts the events actually need does the width grow appreciably. (On the mock as first generated this ladder showed an apparent 18% *improvement* at intermediate depth, $\sigma(H_0)$ falling to 0.82× its complete-catalog value. That gain does not survive the generator repair of §7.4: it was a property of the defective posterior construction interacting with this realisation, not of the survey — a concrete caution against reading structure into single-realisation width ladders.)

The fraction's precision is nearly completeness-independent. What identifies $f_{\rm AGN}$ is the contrast between the two completed tracer priors, and the completion preserves that contrast: as observed hosts vanish, each tracer's prior migrates onto its own smooth $n_{0,k}\,({\rm d}V_c/{\rm d}z)(1+z)^{\delta_k}$ field, and the two fields still differ by the number-density ratio that carried the signal to begin with. The informativeness of the *design* is therefore almost independent of how deep the survey is — as long as $n_{0,k}$ is known, which §6 shows is the whole of the caveat.

Every level stays evaluable. Inadmissible cells fall from 728 to 440 of 3321 along the ladder and never approach the peak, with a posterior mass adjacent to them of at most 2.7×10^{-6}. The selection integral's effective sample size *rises*, from 147×10^3 to 221×10^3, because the incomplete model's target is increasingly dominated by the smooth out-of-catalog term, which the population branch of the proposal covers well. This is the opposite of the failure mode of §4.

5.3. *Is that width real?*

A width that is nearly flat across a sixfold change in completeness invites the suspicion that it is set by something other than the data, so we measured what the same analysis returns when the data cannot inform it. Permuting the per-event sky positions among events preserves every sky patch, every distance and every localisation area, and destroys only the pairing between an event's distance and its own host's redshift. The permuted analysis returns a posterior that is *just as narrow* — 0.93 times the measured width at the complete rung — but centred at $f_{\rm AGN} = 0.1479$ instead of 0.4264, a displacement of 5.21 measured widths (Figure 7b).

Read correctly this validates the measurement rather than undermining it. For a mixture weight the likelihood's curvature is a property of how distinguishable the two host priors are, which is a property of the survey design and not of the realisation; a width that barely responds to the data is what that predicts. What must be data-driven is the *location*, and it is, by nearly five widths. It also supplies the right degradation statistic for the AGN identification itself, which combines width and displacement: the peak-to-null separation falls from 5.21 widths at complete to 4.39 at $C = 0.175$, a loss of 16% (Figure 7c). That is close to the 1.11× width figure and far from the 1.87× one, so it is the fraction's robustness and not the distance scale's that carries over.

5.4. *How to read the centres*

At every rung of the two-tracer ladder, including the complete one, the fraction recovers high — 0.427 down to 0.364 against a planted 0.30 — so it is not a completeness effect. The expansion-rate centres, by contrast, carry no common offset: they scatter from −0.25 to +1.70 $\rm km\,s^{-1}\,Mpc^{-1}$ about the truth along the ladder, consistent with the catalog-realisation term measured in §8 (1.24 $\rm km\,s^{-1}\,Mpc^{-1}$ of scatter per realisation) for a single shared host field. (Before the generator repair the ladder sat uniformly low, with offsets from −1.99 to −0.72 $\rm km\,s^{-1}\,Mpc^{-1}$; the repair removed the common offset and left the realisation scatter.) The high fraction is a change of *estimator* rather than of data: the same mock and the same events under the complete-catalog limit give $f_{\rm AGN} = 0.2629$ (§4), and switching to the completed model with the true per-tracer anchor gives 0.4264, 2.4σ high at the complete rung. The mechanism is specific and worth spelling out, because it is generic to sparse tracers. The completion cannot distinguish "no AGN here" from "no AGN observed here", and with only 154 catalogued AGN inside the horizon the Poisson noise in ${\rm d}N_{\rm obs}/{\rm d}z$ is large enough that Equation (5) returns $C_{\rm AGN}(z) < 1$ in many bins even when the catalog is genuinely complete. A missing-AGN budget is then manufactured out of shot noise and deposited in

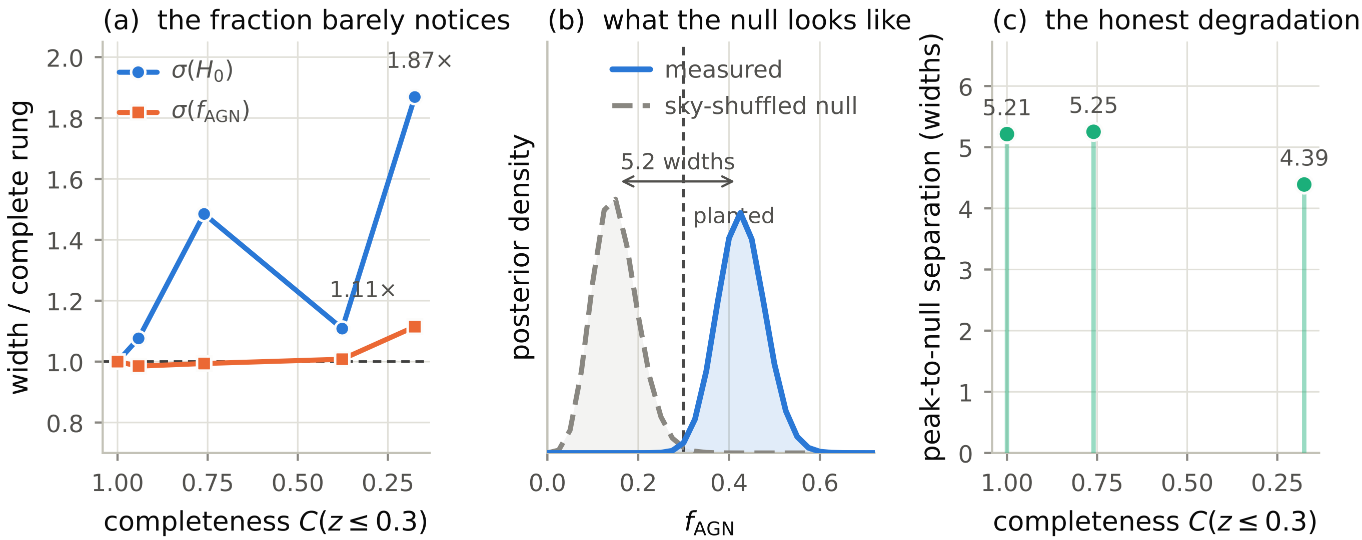

Figure 7. Two-tracer flux-limit ladder at fixed data, with both tracer densities anchored at the truth. *(a)* joint 68% half-widths as factors against the complete-catalog rung: the fraction degrades by 1.11× while the expansion rate degrades by up to 1.87×, non-monotonically, with the first rung costing only 1.08×. *(b)* the measured fraction against its sky-shuffled null at the complete rung; the null is equally narrow but sits 5.21 widths away. *(c)* that separation along the ladder, the degradation statistic for the AGN identification itself.

Table 3. Two-tracer flux-limit ladder at fixed data. C and the host count are for the sparse tracer within $z \leq 0.30$; σ are 68% half-widths from the joint $(H_0, f_{\rm AGN})$ grid, with the factor against the complete rung beside each. The last column is the separation between the measured $f_{\rm AGN}$ peak and its sky-shuffled null, in data widths.

level	C	AGN in horizon	empty (%)	$N_{\rm eff}/10^3$	$\sigma(H_0)$	×	$\sigma(f_{\rm AGN})$	×	separation
complete	1.000	154	37.5	147	0.767	1.00	0.0530	1.00	5.21
$m < 21$	0.942	145	94.5	166	0.826	1.08	0.0522	0.99	···
$m < 20$	0.760	117	97.9	173	1.139	1.48	0.0527	0.99	5.25
$m < 19$	0.377	58	99.4	192	0.850	1.11	0.0534	1.01	···
$m < 18$	0.175	27	99.8	221	1.433	1.87	0.0591	1.11	4.39

empty pixels, where it lets events be AGN-hosted. This is the sparse-tracer form of the degeneracy between the mixture weight and the tracer density, and it is what §6 measures.

6. THE ROBUSTNESS WAS BOUGHT BY KNOWING THE DENSITY

The completeness result of §5.2 was obtained with the sparse tracer's comoving number density pinned at the truth, which is the most favourable case available. It is also the case that makes the mechanism work: the completion converts a *known* n_0 into a missing-host budget, and it is that budget which keeps the two tracer priors distinguishable once their observed hosts are gone. Taking the knowledge away should therefore matter, and it does — more than incompleteness itself.

We map the two-dimensional likelihood in $(f_{\rm AGN}, \log_{10} n_{0,\rm AGN})$ at each rung of the ladder, on a 51 × 201 grid over $\log_{10} n_{0,\rm AGN} \in [-9.6, -7.1]$ with the expansion rate pinned at the truth, and read every level of prior knowledge off that one likelihood as a reweighting rather than a separate run: a delta function at the truth for the anchored case, a Gaussian of width $\log_{10}(1+\epsilon)$ dex for a fractional knowledge ϵ, and a flat prior for no knowledge at all. Nothing else changes — the same mock, the same events, the same catalogs, the same targeted injection sets — so the comparison adds exactly one degree of freedom. No grid cell is inadmissible anywhere in the map.

6.1. *The interaction is the result*

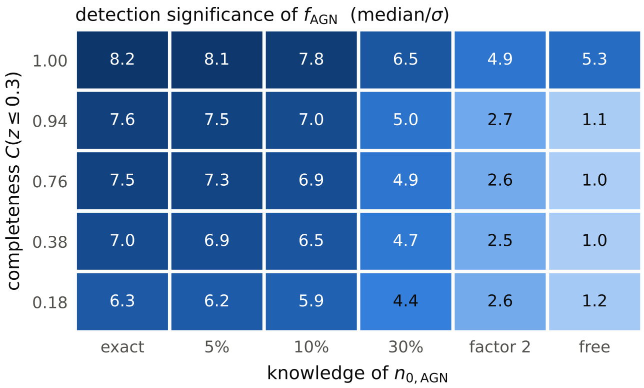

Figure 8. Detection significance of the AGN-hosted fraction, median/σ, across survey completeness (rows) and prior knowledge of the sparse tracer's comoving number density (columns). Every column is a reweighting of the same likelihood grid, so the whole map comes from five grids. The two topmost entries of the free column carry 15.4% and 17.8% of their density posterior against the low edge of the scanned range and are partly statements about that range; the rest of the map is not affected.

Figure 8 gives the detection significance of the AGN-hosted fraction, median/σ, across completeness and density knowledge. With the density known exactly, taking the survey from complete to $C = 0.175$ costs 23% of the significance: 8.2σ becomes 6.3σ, which is the near-immunity of §5.2 seen through a different statistic. With the density free, the same thinning costs 78%: 5.3σ becomes 1.2σ, and almost all of that is spent in the very first step away from a complete catalog, where the significance falls to 1.1σ.

Reading across a row gives what density knowledge buys; reading down a column gives what completeness costs. Neither reading alone is the result — the interaction is. Completeness and density knowledge are not separable axes, because the completion can substitute a modelled budget for missing hosts only if it is told how many hosts there should be.

The practical threshold, though, is encouraging. Knowing $n_{0,\rm AGN}$ to 0.041 dex — about 10% — costs almost nothing at any completeness: 8.2σ becomes 7.8σ at complete, and 6.3σ becomes 5.9σ at $C = 0.175$. A 30% prior costs noticeably more (6.5σ and 4.4σ), and a factor-of-two prior — entirely plausible for a real AGN luminosity function — already sits at 2.6σ once the survey is incomplete. The measurement survives realistic anchoring; it does not survive ignorance.

6.2. *Why the width alone would mislead*

The 68% half-width on $f_{\rm AGN}$ (Table 4) is a poor summary here and would give the wrong answer if quoted on its own. The narrowest entry anywhere in the table is 0.033, and it belongs to the free-density column at the complete rung — which is also among the least significant entries in that row, because the posterior has collapsed onto small $f_{\rm AGN}$ rather than measured it. That is why the significance, not the width, is the headline: a posterior can be narrow and centred anywhere.

6.3. *The degeneracy, and which way it points*

The correlation between the mixture weight and the density under a flat density prior is +0.58 at the complete rung and saturates at +0.90 as soon as the survey thins (Figure 9a). This is the anticipated failure mode made quantitative: a missing AGN host and an AGN-hosted event both explain an event in a pixel with no observed AGN, and once most hosts are missing the two parameters are very nearly the same parameter. The joint region is a curved band running from small $f_{\rm AGN}$ at low density up toward the truth, which at every rung lies at or just beyond the band's upper tip: the flat-prior posterior stops short of it on both axes at once.

The direction of that band explains both of the fraction's biases at once. Under a flat prior the density itself is recovered *below* the truth at every rung — by −1.45, −1.29, −1.17, −0.98 and −0.65 dex, i.e. factors of 27.9 down to 4.4 (Figure 9b). Pinned at the true density, the completion manufactures a missing-AGN budget out of the shot noise of a sparse tracer and pushes $f_{\rm AGN}$ up, which is the high centre of §5.4; allowed to move, it kills that budget by driving the density down and pulls $f_{\rm AGN}$ below the truth instead. The two biases are the same mechanism seen from opposite ends.

That also disposes of a tempting misreading of the map. Following the centre of $f_{\rm AGN}$ across the density priors, the truth falls inside the 68% interval only at intermediate prior widths — the 30% prior on most rungs, the factor-of-two prior at the ends of the ladder — and never in the free column. This is not evidence that an intermediate prior is the right one. The anchored bias is high and the degeneracy pulls low, so *some* intermediate prior width necessarily crosses the truth, and where it crosses is set by the size of the residual bias rather than by anything physical. The honest reading is that the recovered fraction is a strong function of an assumption, over a range of assumptions that are all defensible a priori — which is itself the argument for measuring $n_{0,\rm AGN}$ externally instead of marginalising over it.

6.4. *What this means for the programme*

Three consequences follow, and they are requirements on the *AGN survey* rather than on the gravitational-wave data.

Table 4. 68% half-width on the AGN-hosted fraction across survey completeness (rows) and prior knowledge of the sparse tracer's comoving number density (columns), on the same grids as Figure 8. The width alone is a misleading summary: its smallest entry anywhere in the table belongs to the free-density column at the complete rung, which is among the *least* significant entries of that row.

C	knowledge of $n_{0,\mathrm{AGN}}$					
	exact	5%	10%	30%	factor 2	free
1.00	0.053	0.053	0.054	0.055	0.052	0.033
0.94	0.052	0.052	0.053	0.059	0.059	0.037
0.76	0.052	0.052	0.054	0.060	0.062	0.042
0.38	0.053	0.054	0.055	0.062	0.068	0.053
0.18	0.058	0.059	0.061	0.071	0.087	0.091

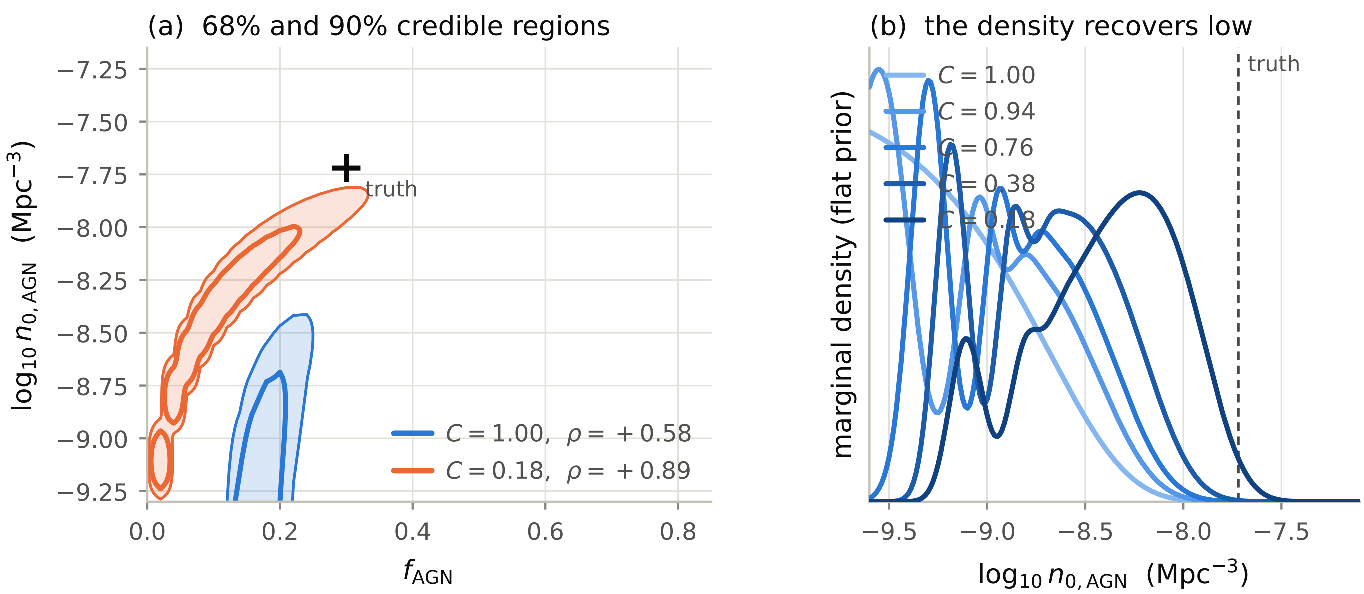

Figure 9. The degeneracy between the AGN-hosted fraction and the AGN comoving number density, with the expansion rate pinned at the truth. *(a)* 68% and 90% credible regions at the complete rung and at $C = 0.175$; the truth (cross) lies at or just beyond the upper tip of the band, which is why an anchored density biases the fraction high and a free one biases it low. *(b)* the flat-prior density marginal at every rung; the density recovers below the truth by 1.45 dex at complete, falling to 0.65 dex at the shallowest rung. At the two shallowest survey depths the marginal keeps 15.4% and 17.8% of its mass against the low edge of the scanned range.

First, anchoring the tracer density to about 10% is the design specification. Below that the measurement degrades gracefully with survey depth; above it, depth stops being the limiting factor.

Second, a fixed-density $\sigma(f_{\mathrm{AGN}})$ must not be quoted as a forecast. At fixed n_0 the width is nearly completeness-independent and reads as a robust measurement; the same configuration with a factor-of-two density prior is a 2.6σ result.

Third, in the incomplete regime the completion's missing-host budget is doing more work than the observed hosts, and its amplitude is set entirely by n_0. Everything downstream of the completion therefore inherits the density's uncertainty, including the completeness-robustness of §5.2.

7. AN ERROR BUDGET FOR THE DISTANCE SCALE

Every recovered expansion rate in this paper is low, on both families of mocks, and the offset does not average down with the number of events. This section reports what it is made of, on the matched single-tracer mock (§2.7). The budget closes: it decomposes into two further respects in which the mock, as generated, was not a draw from the assumed model — each identified, measured, and repaired — and an estimator overhead that is measured against an exact evaluation of the likelihood (§7.4, Figure 10).

7.1. *The offset survives a matched generative process*

Splitting the 1 000-event sample into 10 disjoint blocks — whose scatter measures gravitational-wave realisation noise while a common offset is systematic, since the host catalog and injection campaign are shared and do not average down — gives a mean expansion rate of 66.000 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, i.e. an offset of $-1.74{\pm}0.40$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, or 4.4σ. Reseeding the catalog, the events and the posterior samples together over 5 independent realisations, so that the result is marginalised over catalog realisations rather than conditional on one, gives $-1.61{\pm}0.49$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 3.3σ. The offset is real and it is not a property of one catalog.

It is also not carried by a minority of pathological events. Dividing the sample into 50 disjoint blocks of 20 events, the per-event log-likelihood pull has mean -0.013 with a median absolute deviation of 0.055 and 0 outlying blocks: the pull is spread evenly across the sample, which is what a convention or normalisation error looks like rather than a few bad events.

Several candidate explanations were measured and eliminated. Matching the host rate index to the mock's own host draw moves H_0 by -0.046 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Truncating the catalog to a third or a seventh of its hosts — which moves the redshift boundary a long way while leaving every host the events can use — changes the result by at most 3.9×10^{-4} $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, because at $K=1$ the survey-global normalisation cancels between the per-event terms and the selection term. Coarsening the sky pixelisation moves it by $+0.604$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. And the selection integral itself is right: its logarithmic derivative with respect to H_0 at the truth is 0.0406, matching the analytic expectation for a distance-limited survey. What remains is the per-event term: at the truth the per-event sum alone peaks at 72.14 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $+4.40$ high, and the selection term shifts the total by -5.54 to 66.61 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

7.2. *The offset scales with the measurement noise*

Holding the catalog, the events' true parameters and the injection campaign fixed and varying only the fractional distance uncertainty gives a clean scaling (Figure 11a). With posterior samples drawn about the true parameters the offset is -0.031 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at $\sigma_{d_L}=0.01$, -0.118 at 0.03 and -1.136 at $\sigma_{d_L}=0.10$; with samples drawn about a noisy observation — the construction a real parameter-estimation run produces — it is -0.001, -0.244 and -1.349. Both series vanish as the noise vanishes and grow roughly quadratically in it, which says the offset lives in the measurement-noise channel rather than in the noiseless likelihood. Correcting the posterior construction closes the zero-noise limit essentially exactly but leaves the offset at realistic noise almost unchanged, so it isolates the channel and not the mechanism: a σ^2 scaling is consistent with any defect that scales with the noise level. The defects identified in §7.3 and §7.4 both do.

7.3. *Half of it is a detection rule that acted on the wrong data*

The mechanism we can identify is a mismatch between what decided detection and what the likelihood conditions on. If a source is kept or discarded on the basis of a latent variable that does not appear in the data — a random projection factor entering the signal-to-noise, say, drawn independently of the noise realisation the parameter estimation conditions on — then the correct detected-set likelihood carries an extra factor of $P(\mathrm{det}|\theta)$ *inside* each event's integral,

$$p(\{d_i\}\,|\,\Lambda,\mathrm{det})=\frac{\prod_i\int\mathrm{d}\theta\,p(d_i|\theta)\,p(\theta|\Lambda)\,P(\mathrm{det}|\theta)}{\mu(\Lambda)^{N_{\mathrm{obs}}}},\qquad(7)$$

whereas every population code — correctly, for a real detector — evaluates $\prod_i[\int p(d_i|\theta)p(\theta|\Lambda)\mathrm{d}\theta]/\mu^{N_{\mathrm{obs}}}$, which is exact when detection is a deterministic function of the data so that the indicator drops out. The inference is not wrong; the mock is not a draw from it.

We measured the size of that effect with a paired experiment over 20 realisations in which the two arms share the host catalog, the event seed and every ancillary uncertainty model, so their difference is the detection rule alone (Figure 11b, Table 5). The control arm, with detection decided by the latent projection factor, gives an offset of $-1.570{\pm}0.184$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (8.5σ), reproducing the independent 5-realisation measurement of $-1.610{\pm}0.486$ and licensing the comparison. Moving detection onto a single observed measurement per source, which is then also what the posterior conditions on, gives $-0.802{\pm}0.162$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (5.0σ). The paired difference is $+0.768{\pm}0.226$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, 3.4σ from zero, or 49% of the control offset. The projection latent cannot simply be retained in the corrected arm — that would leave detection depending on a variable absent from the data, which is a different mismatch rather than a fix — so the reference signal-to-noise was recalibrated, to 6.278, to reproduce the control's detected fraction; the two arms' event populations then agree to a few per cent in median redshift, and each arm carries its own injection campaign of 120 million proposals.

After this correction the campaign offset is $-0.80{\pm}0.16$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, still 5.0σ from zero. The next subsection splits it.

7.4. *The rest is the mock again, plus the estimator's overhead*

The generative process of this mock is simple enough that the likelihood it implies can be evaluated *exactly* — the host marginal as a discrete sum over the catalog, the mass integrals and the selection function by deterministic quadrature, no Monte Carlo anywhere — and the exact evaluation splits the residual cleanly. Over the 20 realisations of the campaign, the exact likelihood itself recovers the expansion rate low, by $-0.49{\pm}0.08$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (6.4σ). Equivalently, the score identity that any detected-set likelihood must satisfy at the truth fails: the mean per-event score falls short of the $\mathrm{d}\ln\mu/\mathrm{d}H_0 = 0.0411$ the identity demands by $-0.0048{\pm}0.0007$ per event per $\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Most of the residual is therefore not the inference at all: it is a third respect in which the mock was not a draw from the model, of the same family as the last two. Each event's sky-localisation width was drawn as a deterministic function of its *latent true* masses and distance ($\sigma_{\mathrm{ang}} \propto d_L/\mathcal{M}_c^{5/6}$, the scaling of an optimal signal-to-noise) and then handed to the parameter estimation as known metadata. The width is thereby itself a distance-sensitive — hence H_0-sensitive — observable, and a fixed-width sky posterior cannot represent it: events whose true host received a large sky residual systematically down-weight that host, and the omission has non-zero mean score at the truth. The proof is a paired generator experiment under the exact likelihood: fresh event sets drawn with the recipe as-is give $-0.62{\pm}0.06$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and deriving the width from the *observed* masses and distance — measuring first, then localising — gives $-0.06{\pm}0.07$ and restores the score identity. That is closure.

The three defects state one requirement three ways: for a dark-siren mock to be a draw from the model, every quantity the likelihood conditions on as fixed metadata — the posterior samples' centring, the detection decision, the per-event uncertainty widths — must be a function of the observed data, not of the latent true parameters. Violations do not announce themselves; each of the three produced a smooth, stable, noise-scaled offset of the same sign that survived every conventional robustness test in §7.1–7.2. The score identity at the truth, evaluated under an exact likelihood, is the diagnostic that finds them.

What genuinely belongs to the inference is the paired difference between the production estimator and the exact likelihood on the same events: $-0.31{\pm}0.13$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (2.5σ). Its budget is itself measured (Figure 10): the $1/N_{\mathrm{eff}}$ selection-variance correction (Farr 2019) tilts the posterior by -0.12 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ because N_{eff} rises with H_0; sky pixelation contributes $+0.10{\pm}0.02$, the sample-basis Jacobian $+0.025$, the fixed-width mass kernels -0.037 and the catalog kernel smoothing $+0.015$. On top of these systematic terms the selection estimator carries a zero-mean noise of 0.38 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per realisation (mean -0.02) from the interaction of the shared injection set with each catalog's kernel support — injections drift in and out of contact with individual host kernels as H_0 rescales the distance–redshift map — which is invisible to any single-catalog analysis and is part of why the per-realisation intervals of §7.5 are too narrow. The practical statement for selection budgeting: an effective-sample-size floor guards the mean of the selection estimate but not this H_0-dependent wander, which should be budgeted through the estimator's variance as a function of the scanned parameter rather than at a single point.

The regenerated campaign confirms the attribution end to end on full production mocks. Rebuilding all 20 realisations with the repaired generator and rerunning the production estimator moves the offset from $-0.80{\pm}0.16$ to $-0.39{\pm}0.15$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, a paired per-realisation improvement of $+0.41{\pm}0.04$; what remains agrees with the exact-likelihood attribution realisation by realisation (repaired campaign minus exact likelihood on the original events: $+0.10{\pm}0.12$) and is the estimator overhead of the previous paragraph, not the mock.

7.5. *The host catalog is a random realisation, and it matters*

Independently of any offset, the 5-realisation run measures something that dark-siren error budgets do not usually contain. The realisation-to-realisation standard deviation of the recovered expansion rate is 1.09 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, against a mean quoted 68% half-width of 0.48 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Figure 12). The credible intervals are therefore too narrow by a factor of 2.2, and quadrature-subtracting them from the realised scatter leaves 0.97 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ of scatter per 1 000-event sam-

Table 5. Error budget of the distance scale on the matched single-tracer mock. The first row is the offset the mock as generated produces; the second is the paired difference between two arms that differ only in the detection rule; the third is the offset of the exact likelihood itself, which isolates the latent sky-width defect; the fourth is the paired difference between the production estimator and the exact likelihood on the same events. The last row is the closure endpoint: the exact likelihood on events drawn with the repaired recipe.

	$N_{\rm real}$	ΔH_0 $(\mathrm{km\,s^{-1}\,Mpc^{-1}})$	σ
offset as generated	20	-1.570 ± 0.184	8.5
detection acting on latent data	20	$+0.768 \pm 0.226$	3.4
sky width drawn from latent parameters	20	-0.489 ± 0.077	6.4
estimator overhead (vs. exact likelihood)	20	-0.312 ± 0.125	2.5
closure: exact likelihood, repaired recipe	20	-0.062 ± 0.066	0.9

Note—Independent of the offset, the host-catalog realisation contributes $0.97\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ of scatter per 1000-event realisation, which the per-realisation intervals do not contain: the realised scatter is $2.2\times$ the quoted 68% half-width.

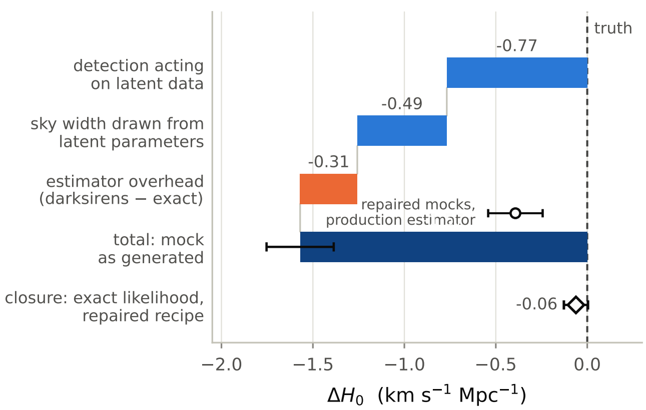

Figure 10. The distance-scale budget closes. From the campaign offset of the mock as generated, the two generator defects — detection acting on latent data (§7.3) and a sky-localisation width derived from latent parameters (§7.4) — account for their measured shares, leaving the estimator-minus-exact-likelihood overhead, whose own budget is the Farr $1/N_{\rm eff}$ term plus zero-mean selection-Monte-Carlo noise. The open circle is the regenerated campaign — the production estimator on repaired-generator mocks — which lands on the overhead, as the attribution predicts; the open diamond is the closure endpoint, the exact likelihood on events drawn with the repaired recipe.

ple attributable to the host catalog realisation itself, at this catalog density.

The reason is structural rather than incidental. A credible interval computed with one catalog is conditional on that catalog: it propagates the gravitational-wave measurement uncertainty and the selection estimate, but the catalog enters as a fixed object, so the fact that a different realisation of the same density field would have placed hosts differently is nowhere in it. At this density the missing term is comparable in size to the offset of §7.3, so it is not a refinement. Any dark-siren analysis that quotes an interval at fixed catalog — which is to say, any analysis using a single real catalog — is missing a term of this character, and its size has to be established by realisation rather than assumed small.

8. CATALOG REALISATION AND THE TWO-TRACER MEASUREMENT

The two-tracer numbers of §4 and §5.2 come from a single draw of the host field, and §7.5 shows that for a single tracer the realisation term is comparable to the statistical one. Whether the multitracer combination cancels part of it is a different question, and an interesting one: the two tracers sample the same underlying density field, so a realisation that happens to over-populate a structure over-populates it in both, and the *contrast* the fraction is measured from may be more stable than either prior separately.

We measure this by rebuilding the matched two-tracer mock — host field, tracer subset, events, posterior samples and targeted injection campaign — over 12 independent realisations and repeating the fraction scan and the joint scan on each. These realisations were generated before the sky-width repair of §7.4, so their absolute expansion-rate offsets carry that known defect in full; the realisation-to-realisation *scatter*, and its ratio to the quoted widths, is the estimand here, and the comparison between the fraction and the distance scale is internal to the ensemble.

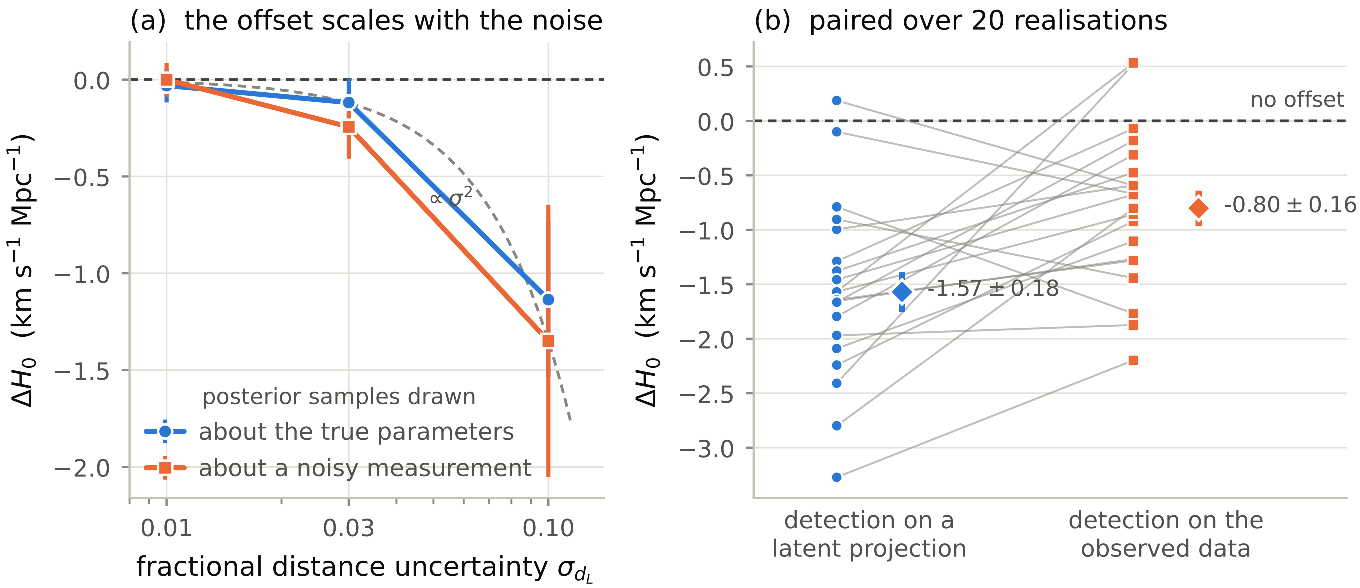

Figure 11. Error budget of the distance scale on the matched single-tracer mock. *(a)* the offset against the fractional distance uncertainty, for posterior samples drawn about the true parameters and about a noisy observation, with a quadratic reference anchored on the corrected arm; both series vanish with the noise. *(b)* the paired detection-rule experiment over 20 realisations. Thin lines join the two arms of the same realisation; diamonds are the arm means with their standard errors. Moving detection onto the data the posterior conditions on removes 49% of the offset and leaves 0.80±0.16 km s^{-1} Mpc^{-1}, which §7.4 splits between a third generator defect and the estimator's overhead.

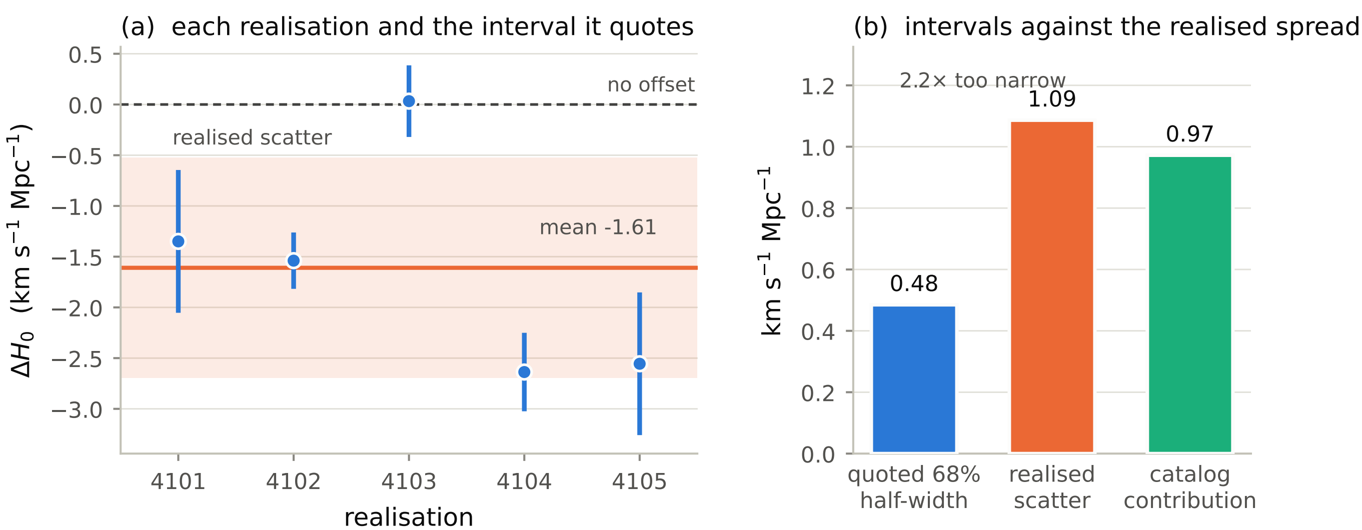

Figure 12. Host-catalog sample variance. *(a)* 5 realisations in which the catalog, the events and the posterior samples are reseeded together, each with the 68% interval it quotes; the band is the realised scatter about the mean. *(b)* the same statement as a budget: the realised scatter exceeds the quoted half-width by a factor of 2.2, and the excess in quadrature, 0.97 km s^{-1} Mpc^{-1}, is the catalog realisation's own contribution.

The answer is that the contrast is realisation-stable and the distance scale is not. At fixed true expansion rate, the fraction's mean offset over realisations is -0.054 ± 0.009 (5.9σ from zero) with a scatter of 0.031 against a mean quoted 68% half-width of 0.039: a realisation factor of 0.8, i.e. the quoted intervals for $f_{\rm AGN}$ are honest about the catalog draw. On the joint grid the same statement holds — offset -0.027 ± 0.013, scatter 0.045 against 0.042, factor 1.1. The expansion rate is the opposite: a mean offset of -3.22 ± 0.55 km s^{-1} Mpc^{-1} — dominated by the pre-repair generator defect — with a scatter of 1.91 km s^{-1} Mpc^{-1} against a mean quoted half-width of 0.72, a factor of 2.7. The single-tracer factor of 2.2 (§7.5) is therefore not softened by the second tracer for the distance scale, but the parameter the second tracer exists to measure is realisation-stable at the quoted precision: the shared density field drops out of the contrast, as the opening paragraph hoped, while H_0 keeps the full realisation term of whichever priors it leans on.

The 1.8σ-low pre-repair fraction of §4 is consistent with this ensemble: at scatter-over-width near unity, single-draw offsets of that size occur with ordinary probability. The ensemble does show a small mean deficit — -0.027 ± 0.013 (2.1σ) on the joint grid and -0.054 ± 0.009 (5.9σ) at fixed H_0 — which demands re-examination on a repaired ensemble, since at expansion-rate offsets this large the distance scale and the fraction are not fully decoupled. That ensemble follows.

Regenerating the same 12 realisations with the repaired generator of §7.4 settles both points. The expansion rate closes: the post-repair ensemble recovers H_0 at $+0.44\pm0.36$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (1.2σ from zero), so the two-tracer joint measurement is unbiased once the mock is a genuine draw from the model the likelihood assumes, and what remains of the realisation term — scatter 1.24 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ against a quoted half-width of 0.86, a factor of 1.4 — is the honest catalog-draw contribution rather than a defect. The fraction's small deficit survives at reduced size — -0.021 ± 0.008 (2.7σ) on the joint grid, with realisation factor 0.7 — which places it at the same scale as the residual estimator terms of §7.4 rather than at the distance-coupled size it had before the repair. The fraction remains realisation-stable throughout; the distance scale becomes accurate as well as stable.

9. DISCUSSION

9.1. *What a sparse tracer actually adds*

The result of §3 is that the mixture weight between two host priors is measurable, at the few-per-cent level from a thousand dark sirens, over its entire range. The result of §6 is that this is a statement about a survey pair and not about the gravitational-wave data: what the sparse tracer contributes is a sky-and-redshift density contrast against the dense one, and the contrast is only as well determined as the sparse tracer's number density. In that sense the AGN-hosted fraction behaves less like a gravitational-wave population parameter and more like a cross-correlation amplitude, with the usual consequence that the tracer's own selection function enters at leading order.

That framing also places the measurement in the literature. Existing constraints on the AGN channel's contribution come from population and rate arguments — comparing observed masses, spins and rates against AGN-channel predictions (Gayathri et al. 2021; Ford & McKernan 2022; Abbott & et al. 2023b) — or from seeking direct spatial association of individual events with catalogued AGN and flares (Graham et al. 2020; Bartos et al. 2017), and AGN have been proposed as dark-siren host catalogs in their own right (Bom & Palmese 2024). The mixture weight measured here is a different quantity from either. Unlike a population argument it uses no mass or spin information at all — by construction f_{AGN} enters only the host prior — and unlike a per-event association it requires no individual host identification; it is closest in spirit to a cross-correlation amplitude between the event set and the AGN density field, measured relative to the same correlation with the galaxies. The distinction matters for combining them: a mass-spectrum f_{AGN} and a host-prior f_{AGN} are nearly independent measurements of the same branching ratio, and their agreement or disagreement is itself a test of the AGN channel's assembly physics.

9.2. *Requirements on a real AGN survey*

Two requirements follow from the measurements above, and they trade against each other in a way that is not obvious from either alone. Depth is nearly free: from complete to 17.5% completeness within the horizon the fraction's precision degrades by 1.11× and its detection significance by 16%, so an AGN survey need not be deep to be useful. Knowledge of the number density is not free: a factor-of-two prior on $n_{0,\mathrm{AGN}}$ turns a 8.2σ measurement into a 2.6σ one once the survey is incomplete, while about 10% knowledge costs almost nothing. The design conclusion is therefore to spend effort on the completeness *model* — on knowing how many AGN there should be, per unit volume, as a function of redshift — rather than on finding more of them.

Whether 10% knowledge is available now depends on which population the density is required for, and the distinction is the useful conclusion. For a *survey-defined* AGN class the census largely exists at these redshifts: ultra-hard X-ray selection is nearly blind to obscuration, and the Swift-BAT-based luminosity, black-hole-mass and Eddington-ratio distribution functions at $0.01 \leq z \leq 0.3$ pin the sample's number-count normalisation at the few-per-cent level statistically (Ananna et al. 2022). Since the mixture likelihood only ever needs the space density of the population its catalog actually contains, an $n_{0,\mathrm{AGN}}$ good to $\sim$10% for a fixed flux-limited class, over the horizons considered here, is within reach of existing data. What is not known to 10% — or to a factor of a few — is which AGN class hosts the mergers: rate predictions for the channel span orders of magnitude in large part because the assumed hostdisk population does, from Seyfert-like number densities down to quasar ones (McKernan et al. 2018). The requirement of §6 is therefore realistic in exactly the form the analysis needs it — a density defined by the survey that supplies the catalog — while the choice of which surveyed class to point the mixture at remains a physics input, the same one the population-based constraints are trying to measure (Gayathri et al. 2021; Ford & McKernan 2022).

9.3. *The weak depth preference*

The near-flatness of the two-tracer ladder at its shallow end (§5.2) has a simple cause — hosts beyond the gravitational-wave horizon contribute prior weight at redshifts no event can occupy, so the first hosts a flux limit removes are hosts the analysis never needed — and it should generalise to any catalog-based analysis whose horizon is well inside the catalog's depth. The practical statement for survey design is that depth beyond the horizon buys little and a depth cut matched to the horizon costs little. We do not claim the stronger version — that such a cut *sharpens* the measurement: the apparent intermediate-depth improvement on the mock as first generated did not survive the generator repair (§5.2), and the flatness itself is the design-relevant result.

The caveat is that the ladder here imposes an isotropic flux limit. A real survey is anisotropic, and an anisotropic completeness modelled as isotropic imprints a spurious sky-density contrast — which is exactly the channel that identifies $f_{\rm AGN}$. That stress test is the natural next step and we have not done it.

9.4. *What it takes for a mock to be a draw from the model*

The distance-scale offsets in this paper turned out to be, in every measured instance, statements about consistency between a generative process and a likelihood rather than about the estimator's numerics — and the specific inconsistencies found are ones that real mock campaigns can and do contain. On the clustered mocks, a detection cut applied in a variable the selection integral does not use created two opposing multi-$\mathrm{km\,s^{-1}\,Mpc^{-1}}$ levers whose imperfect cancellation looked like a smooth, stable cosmology bias (§4.2). On the matched mocks, three versions of the same violation — posterior samples centred on the truth, a detection decision taken on latent data, and per-event metadata (the sky width) derived from latent parameters — each produced offsets of the same sign and character (§7). None of them was found by robustness testing: each survived depth cuts, pixelisation changes, rate-index changes and event-subset splits. What found them was demanding closure in expectation — the score identity of the detected-set likelihood at the truth — under an exact evaluation of the likelihood the mock implies. We suggest that as the standard of proof a dark-siren mock campaign should meet: not that the recovered values look right, but that every quantity the likelihood conditions on as metadata is a function of the observed data, and that the exact likelihood's score at the truth vanishes.

What then remains chargeable to the inference itself is the estimator overhead of §7.4, -0.31 ± 0.13 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at this configuration, and its budget carries two practical lessons. The systematic part is dominated by the $1/N_{\rm eff}$ selection-variance correction (-0.12 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$), which tilts the posterior whenever $N_{\rm eff}$ depends on the scanned parameter; for grid evaluations, where the correction's premise of a parameter-independent variance penalty does not hold, it should either be removed or budgeted explicitly. The stochastic part — 0.38 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ per realisation from the injection-set–catalog-kernel interaction — does not average down within one analysis, is invisible at a single parameter point, and argues for budgeting the selection estimator by its variance profile across the scan rather than by an effective-sample-size floor at the peak.

9.5. *Limitations*

Beyond the estimator overhead, four limitations bound what the numbers above support.

The clustered mocks' detection rule. The flagship $K = 2$ mocks (§2.7) select events by a cut on true source redshift. No detector can do that, and no injection campaign recorded in $(m_1^{\rm det}, q, d_L)$ can represent such a region, because a fixed region in source-frame redshift is not a fixed region in the observables. Section 4.2 quantifies the consequence exactly — two opposing levers on the distance scale, fully accounted for and removed by a repaired estimator — so the clustered mocks' H_0 values are understood rather than mysterious. But the production scans of §3 were run with the standard estimator, not the repaired one, so those clustered measurements are not closure tests and should not be read as such; their $f_{\rm AGN}$ results stand because the mixture weight is insensitive to the levers.

The matched mocks are unclustered. Conversely, the completeness and density results come from mocks in which the tracers are uniform in comoving volume, so $f_{\rm AGN}$ there is identified by the number-density contrast alone. The bias contrast of the clustered mocks should only help, but we have not measured the completeness ladder with it.

Fixed population and cosmology. The mass and spin population is held at the mock's truth and $\Omega_{m,0}$ is pinned, so the mass distribution is informative but not marginalised. The spectral-siren information this leaves in the likelihood couples the detector-frame masses to H_0, and a fully marginalised analysis will have wider intervals than those quoted here.

One catalog per configuration, except where stated. The completeness and density ladders use one host-field realisation, so their per-level precision is a single realisation and §7.5's realisation term applies to them too. The 12-realisation ensemble of §8 bounds the exposure: the fraction's realisation scatter is comparable to its quoted width (factor 0.7 at fixed H_0 on the repaired ensemble), so the ladder's $f_{\rm AGN}$ comparisons — differences of order several widths — are not realisation-limited, while any H_0 statement at fixed catalog carries a realisation term of order the quoted interval itself (factor 1.4).

10. CONCLUSIONS

We have asked what a sparse, strongly clustered second tracer adds to a dark-siren analysis that already has a dense galaxy catalog, and what a real survey would have to deliver for the addition to be worth making. Working on clustered lognormal mocks with a galaxy–AGN bias contrast of 1.67 and a number-density ratio of 100, and on mocks whose generative process matches the inference exactly, we find:

1. **The AGN-hosted fraction is measurable across its whole range.** From 1 000 dark sirens on complete clustered catalogs, $f_{\rm AGN}$ is recovered with the planted value inside the 68% interval at every interior point, residual offsets no larger than 0.025, and 68% half-widths of 0.016–0.025. Jointly with the expansion rate the two parameters are nearly uncorrelated at $f_{\rm AGN} = 0.307$ ($\rho = -0.01$), so the mixture weight is measured essentially independently of the distance scale.

2. **A sparse tracer needs a selection campaign that follows it.** With injections drawn from the source population, the catalog-conditioned selection integral's effective sample size falls by a factor of 66 across the $f_{\rm AGN}$ grid and the posterior peaks against the edge of the usable region, giving an apparent recovery that carries no information. A proposal that places 0.25 of its draws at catalogued AGN reverses that trend, by a factor of 30, at a cost of 20% of the effective sample size when no events are AGN-hosted — and changes the recovered fraction, not merely its error bar.

3. **Incompleteness is cheap for the fraction and not for the distance scale.** From complete to 17.5% completeness within the horizon — catalogued AGN inside the horizon falling from 154 to 27 — $\sigma(f_{\rm AGN})$ grows by 1.11× while $\sigma(H_0)$ grows by up to 1.87×, non-monotonically and with the first rung costing only 1.08×, because the hosts a flux limit removes first lie beyond the gravitational-wave horizon. The separation between the measured fraction and its sky-shuffled null falls only from 5.21 to 4.39 widths, a 16% loss.

4. **That robustness is bought entirely by knowing the tracer density.** Freeing $n_{0,\rm AGN}$ opens a degeneracy with the mixture weight that reaches $\rho = +0.90$, and the fraction's detection significance falls from 8.2σ to 1.2σ across the same ladder. Completeness and density knowledge are not separable axes. The practical specification is external knowledge of $n_{0,\rm AGN}$ to about 10%, which costs almost nothing at any completeness; a factor-of-two prior already gives only 2.6σ once the survey is incomplete.

5. **The clustered mocks' distance-scale deficit is a selection consistency effect, and the sparse tracer damps it.** The deficit is the incomplete cancellation of two levers sprung by one inconsistency — a true-redshift detection cut against a distance-space selection and a host prior with support above the horizon: −4.01 to −4.27 km s^{-1} Mpc^{-1} from the selection slope (a zero-parameter analytic model reproduces 96% of it) opposed by +3.72 to +1.19 of numerator pull from above-horizon catalog support. An estimator matched to the detection rule recovers the truth at both planted fractions. The deficit grows with $f_{\rm AGN}$ because the sparse tracer's near-delta redshift kernels anchor the per-event term like a noisy counterpart, collapsing the numerator lever — a genuinely multitracer robustness that dense smooth catalogs do not have.

6. **The matched-mock error budget closes, and it indicts the mock before the inference.** H_0 comes back low by 1.57±0.184 km s^{-1} Mpc^{-1} (8.5σ) as generated. A detection rule acting on a latent variable absent from the data accounts for +0.768±0.226; an exact evaluation of the likelihood attributes −0.49±0.08 more to a

sky-localisation width drawn from latent parameters, a defect that biases even the exact likelihood (6.4σ) because the width is itself an H_0-sensitive observable. With both repaired the exact likelihood closes (-0.06 ± 0.07), and the full campaign regenerated with the repaired generator lands on the estimator overhead as predicted (-0.39 ± 0.15). What is chargeable to the estimator is -0.31 ± 0.13: the $1/N_{\rm eff}$ selection correction (-0.12) plus a zero-mean selection-Monte-Carlo wander of 0.38 $\rm km\,s^{-1}\,Mpc^{-1}$ per realisation. Every matched-mock campaign in this paper was regenerated and rerun on the repaired generator, so the recoveries quoted above are absolute, with that estimator overhead as their measured floor.

7. **The host catalog's own realisation is a missing error term.** Over 5 realisations in which catalog, events and posterior samples are reseeded together, the recovered expansion rate scatters by 1.09 $\rm km\,s^{-1}\,Mpc^{-1}$ against a mean quoted 68% half-width of 0.48 $\rm km\,s^{-1}\,Mpc^{-1}$: the intervals are too narrow by a factor of 2.2, and 0.97 $\rm km\,s^{-1}\,Mpc^{-1}$ of scatter per 1 000-event sample is attributable to the catalog realisation. This term is absent from intervals computed at fixed catalog, and at this density it is comparable in size to the systematic above.

Taken together these results say that a two-tracer dark-siren analysis can measure the AGN-hosted fraction, that survey depth is not what limits it, and that the limiting external input is the sparse tracer's space density. They also say that the field's error budgets are currently missing a catalog-realisation term of order 0.97 $\rm km\,s^{-1}\,Mpc^{-1}$ per thousand events; and they suggest a standard of proof for mock-based closure tests — the score identity at the truth, checked under an exact evaluation of the likelihood — since every offset in this paper that looked like an inference systematic turned out, on that test, to be a mock that was not a draw from its own model, plus a measured and budgeted estimator overhead.

[**TODO: Add an acknowledgements block, a software section, and a data-availability statement pointing at the reproduction scripts, once the reference list and the release plan exist.**]

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